Camera-Based Handwashing Assessment

Visual Recognition, Temporal Segmentation
and Adaptive Action Confirmation

Group 43

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Abstract

This project develops a local camera system that recognizes six handwashing actions, reconstructs their order and assesses recognized washing duration. Correct frame labels alone are insufficient for this task: transitions, occlusion and hand exchanges can introduce short erroneous runs that become extra steps in a procedure. Conversely, an accurately recognized sequence may faithfully describe an incorrect washing order. The system therefore separates visual recognition, temporal interpretation and procedural assessment.

An initial PSKUS feasibility study motivates temporal context and boundary supervision. Target-scene development then uses a self-recorded corpus containing 6,987 sampled frames with annotation records, including standard sequences, reordered actions, omissions and isolated gestures. The proposed pipeline combines YOLO features, a unidirectional gated recurrent unit, learned boundary evidence and adaptive action confirmation. Segment duration is assessed relative to the recording and jointly with confidence, replacing a fixed three-second confirmation rule without forcing a standard order.

Four complete pipelines are compared on 632 exactly aligned valid centers from ten reviewed videos. The proposed method achieves 96.52% accuracy, 93.71% macro F1 and 94.40% output-boundary F1. Its final action chains match all ten references, including nonstandard sequences. VideoMAE achieves the strongest classification and boundary scores and the same final-chain result. The proposed model uses approximately 10.64 million parameters, compared with 86.23 million for VideoMAE, and supports recurrent local feedback. The camera application runs one selected model per session, caches predictions and finalizes the action chain and duration after pending work completes. The report connects these results to annotation quality, segmentation behavior and deployment requirements, and distinguishes retrospective internal evidence from future independent validation.

Keywords: handwashing recognition; temporal segmentation; YOLO; GRU; boundary awareness; adaptive confirmation; local camera assessment.

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1 Introduction and System Objectives

1.1 From recognizing a gesture to assessing a procedure

A camera-based handwashing system must answer several different questions. It must identify the action visible at a particular time, determine when one action changes to another, reconstruct the sequence performed across the recording, and establish whether enough washing time has been observed. These questions are related, but they are not interchangeable. A high proportion of correctly classified frames can coexist with an incorrect action chain if a few brief mistakes introduce extra steps. Similarly, a correctly reconstructed nonstandard sequence is a recognition success even though the procedure does not satisfy the application's completion rule.

The project addresses this distinction through an explicit processing pipeline. Images supply evidence about hand configuration. A temporal model relates the current image to the recent sequence. Boundary modeling proposes divisions between action intervals. Adaptive confirmation evaluates whether candidate segments are sufficiently supported to enter the reported chain. Finally, a separate assessment layer compares that chain and the accumulated recognized washing time with the application's requirements. Figure 1 presents these stages as a user-facing system rather than as a classifier alone.

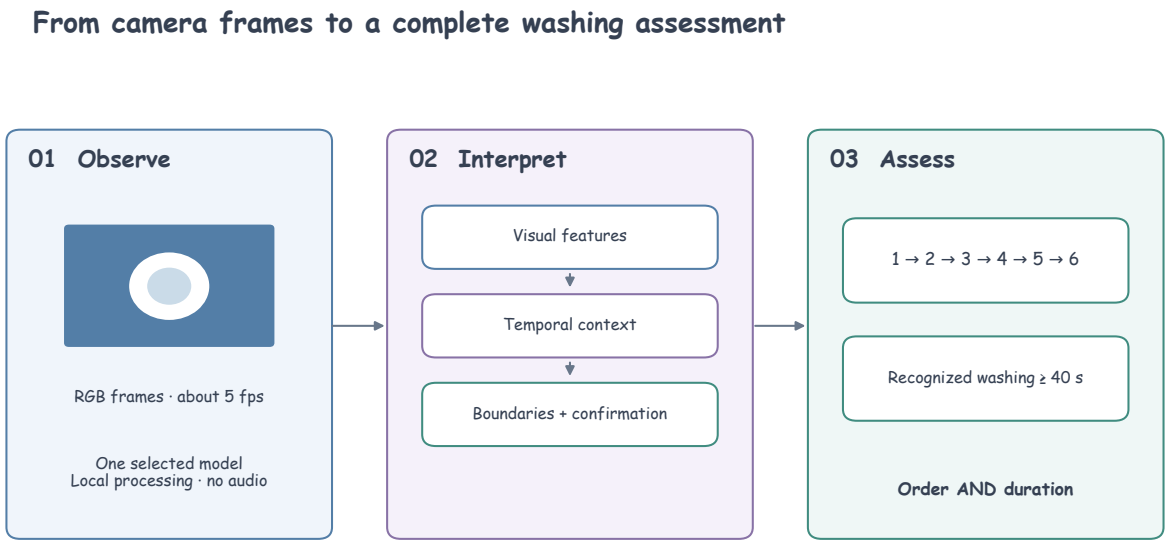


Figure 1: The observation, interpretation and assessment stages. The standard six-step chain is checked after recognition; it is not imposed as a decoding constraint.

This separation has a practical consequence. When an action is missing or repeated, the recognizer should preserve that evidence. An algorithm that always returns the standard order would make the interface look stable but would remove the information needed to assess incorrect procedures. The final chain must therefore remain capable of describing omissions, reversals and repetitions. Stability is useful only when it is achieved without replacing the observed procedure with an expected one.

1.2 Design objectives and delivered contribution

The intended setting is a locally operated camera dashboard. A user selects a model, positions both hands in view, starts a session and receives provisional feedback while washing. After the session ends, the application reports the recognized sequence and washing duration. One model runs at a time, allowing the selected method to have a clear computational and interpretive role. The system does not collect audio, and the retained session evidence is stored locally.

Three objectives shape the design. First, the system should distinguish the six target gestures and background activity using the self-recorded scene. Second, it should convert noisy temporal predictions into an inspectable action chain without assuming the correct order. Third, the final assessment should be available from predictions accumulated during the session, avoiding an unnecessary second pass over all images. These objectives motivate a compact recurrent model and explicit segment-level reasoning.

The primary model is YOLO + Adaptive Boundary-Aware GRU. Its overall architecture is introduced early in Figure 2, so that the later data, method and deployment discussions can refer to a common design. YOLO extracts visual features, a unidirectional GRU encodes temporal context, and two prediction branches provide class and boundary evidence. The downstream confirmation stage combines the relative duration of a segment with its confidence. The system contribution lies in integrating these elements with annotation-aware evaluation and a working local feedback application. It does not require every constituent operation to be a newly invented neural-network primitive.

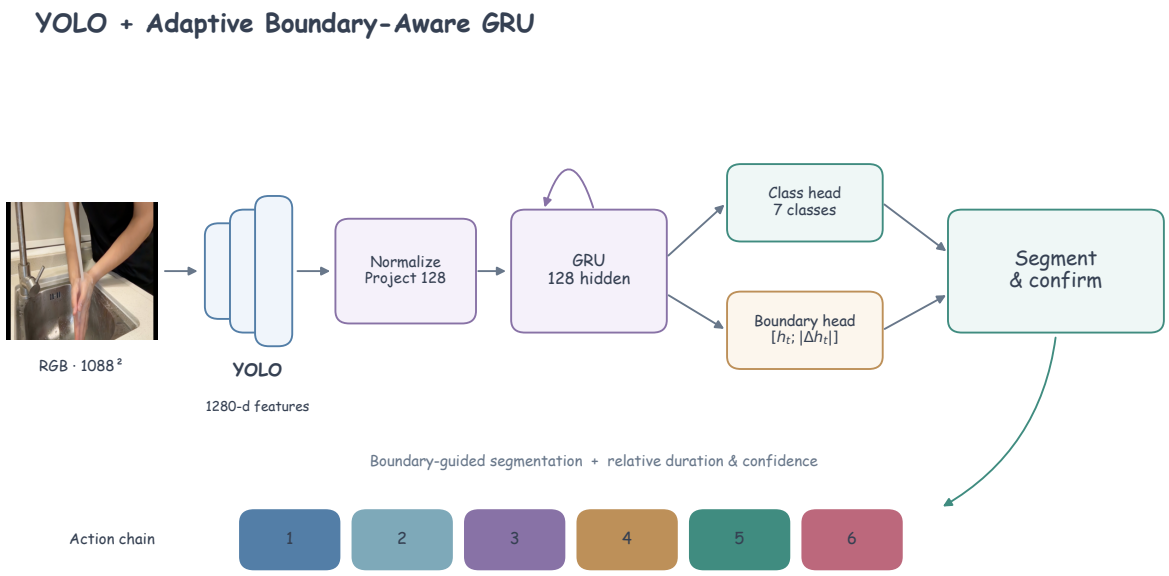


Figure 2: The proposed architecture. Frozen YOLO features feed a projected recurrent representation, class and boundary heads, and adaptive segment confirmation. The input thumbnail is a real scene image; the illustrated output sequence is schematic.

The comparison includes three alternatives with distinct purposes. Standalone YOLO establishes what image appearance can achieve. YOLO + GRU isolates the practical value of recurrent context within the YOLO family. VideoMAE supplies a larger video-transformer accuracy reference. The final comparison concerns the complete configured systems, because their input sizes, pretraining,

final fitting procedures and chain processing differ. These differences are documented rather than hidden behind a single ranking.

1.3 Scope and report structure

The delivered system observes visible actions and recognized duration. It does not measure microbial cleanliness, soap distribution over the skin, applied pressure or the physical quality of contact. A confident action label is evidence about the model's classification, not direct proof of hygiene effectiveness. This distinction keeps the application's completion verdict aligned with what its sensors and labels can support.

The report follows the development logic. It first establishes the action vocabulary and preliminary public-data evidence, then explains the target-scene corpus and its annotation. The middle sections derive the visual, recurrent, boundary and confirmation components. Training and evaluation are described before the results, allowing the reader to interpret the figures under the correct protocol. Case studies connect aggregate metrics to actual timelines. The final sections examine local deployment, limitations and reproducibility. Detailed sequence outputs and implementation contracts are retained in the appendices for inspection.

2 Task Context and Preliminary Feasibility

2.1 The six-action vocabulary

The application uses seven classes: Other and six handwashing gestures. The gestures distinguish palm-to-palm contact, rubbing the hand dorsum, interlaced palms, rubbing the backs of fingers, rotational thumb rubbing and fingertip-to-palm contact. Figure 3 shows representative images from the target scene. A consistent class vocabulary is necessary across annotation files, checkpoints, probability vectors, interface labels and evaluation code. A permutation of otherwise plausible labels would corrupt the final order even if the underlying visual features were useful.

The WHO handwashing poster describes a complete procedure lasting 40–60 seconds, including wetting, applying soap, the rubbing gestures, rinsing and drying [1]. The present classifier observes the six rubbing actions plus Other. Consequently, the application's threshold of 40 seconds of recognized washing is a project-specific assessment rule; it is not a direct implementation of the WHO's full-procedure timing definition. Making this distinction avoids interpreting unobserved rinsing or drying as measured washing evidence.

The annotation vocabulary also distinguishes a hand exchange from a change of action. Repeating the same rubbing gesture with the opposite hand remains the same action class. By contrast, an interval in which the hands rearrange and the gesture is unclear may be marked as an uncertain transition. Table 1 defines the retained class meanings. The uncertain status is represented by a validity flag, not by adding another action to the seven-class output.

2.2 Why temporal models are relevant

An image classifier assesses each observed hand configuration independently. It can recognize distinctive appearances well, but adjacent frames may receive different labels when contact is occluded or the hands rotate. A recurrent representation provides a mechanism for relating successive observations. The GRU formulation introduced by Cho and colleagues uses learned gating within a

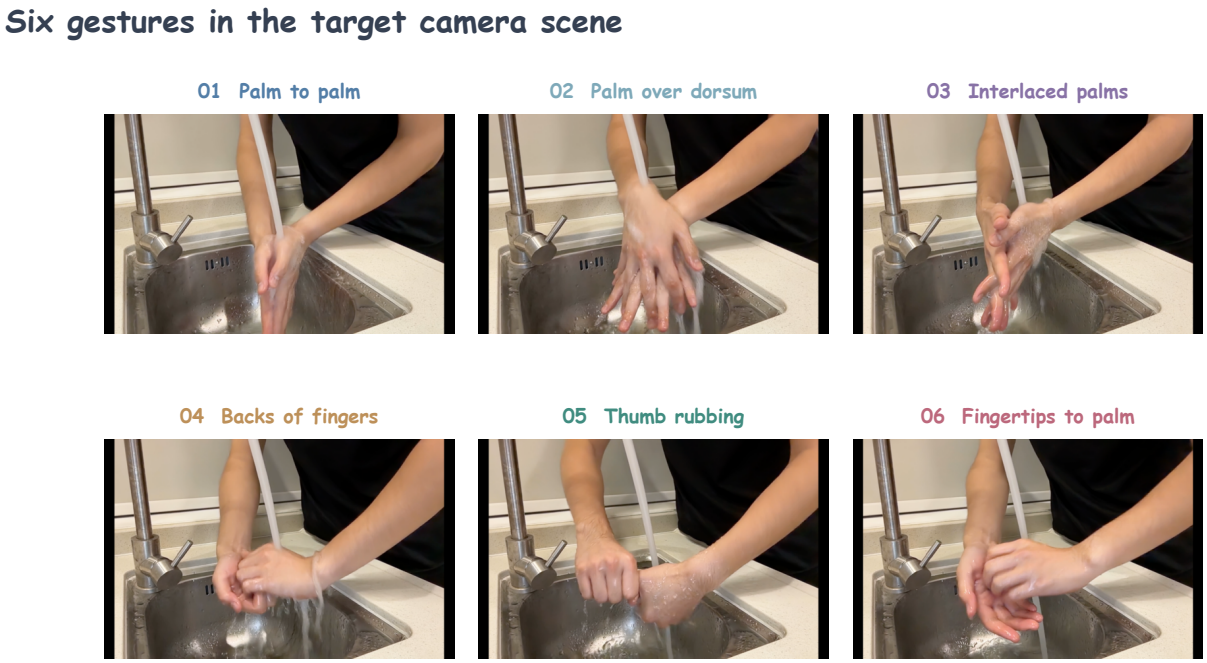


Figure 3: Representative target-scene images for the six action labels. These user-supplied screenshots illustrate the vocabulary and camera view.

Table 1: Seven-class action vocabulary

ID	Action	Annotation interpretation
0	Other	Background or non-washing activity
1	Palm to palm	Rub the two palms together
2	Palm over dorsum	Rub one palm over the opposite hand dorsum
3	Interlaced palms	Rub palms with fingers interlaced
4	Backs of fingers	Rub the backs of fingers against the opposing palm
5	Thumb rubbing	Rotational rubbing of each thumb
6	Fingertips to palm	Rotational rubbing of fingertips against the palm

recurrent encoder [3]. In this project, the relevant design property is that the hidden state can be updated incrementally as new image features arrive; no language-model interpretation is implied.

Temporal action segmentation also requires decisions about the number and extent of action instances. The MS-TCN work explicitly addresses over-segmentation using a multi-stage temporal model and smoothing loss [5]. It is included here as methodological context, not as an evaluated competitor. The present project uses a recurrent model with boundary supervision and a segment-confirmation stage. Its evaluation similarly needs to distinguish frame recognition from the quality of the resulting temporal structure.

VideoMAE provides a complementary representation strategy. Its original work develops masked video reconstruction for representation learning and reports that masking temporally redundant video content can yield effective video representations [4]. The project uses a generic Kinetics-pretrained VideoMAE checkpoint and fine-tunes it for the target labels. It does not reproduce the original self-supervised pretraining experiment. The model's larger video representation makes it

a useful accuracy reference, while its window-based input creates a different deployment pattern from incremental recurrent inference.

2.3 PSKUS as an initial feasibility study

The PSKUS resource contains 3,185 handwashing episodes recorded in Pauls Stradins Clinical University Hospital, with frame-level movement annotations and multiple annotators [2]. The project's preliminary experiment used 615 videos from DataSet1–4, divided into 430 training, 92 validation and 93 test recordings. It sampled at 5 fps and compared aligned centers of 16-frame windows with a stride of four samples. Agreement between the two designated annotators yielded 5,599 evaluable centers, including 1,068 near transitions [6].

Figure 4 summarizes the historical feasibility results. Ordinary GRU increased overall macro F1 from 90.25% to 92.71%, while the near-transition score increased from 83.27% to 90.06%. The early boundary-aware model reached 94.06% overall macro F1 and 91.01% near-transition macro F1. These findings motivated temporal context and attention to transitions. They are evidence from the preliminary experiment and should not be combined numerically with the later self-recorded comparison.

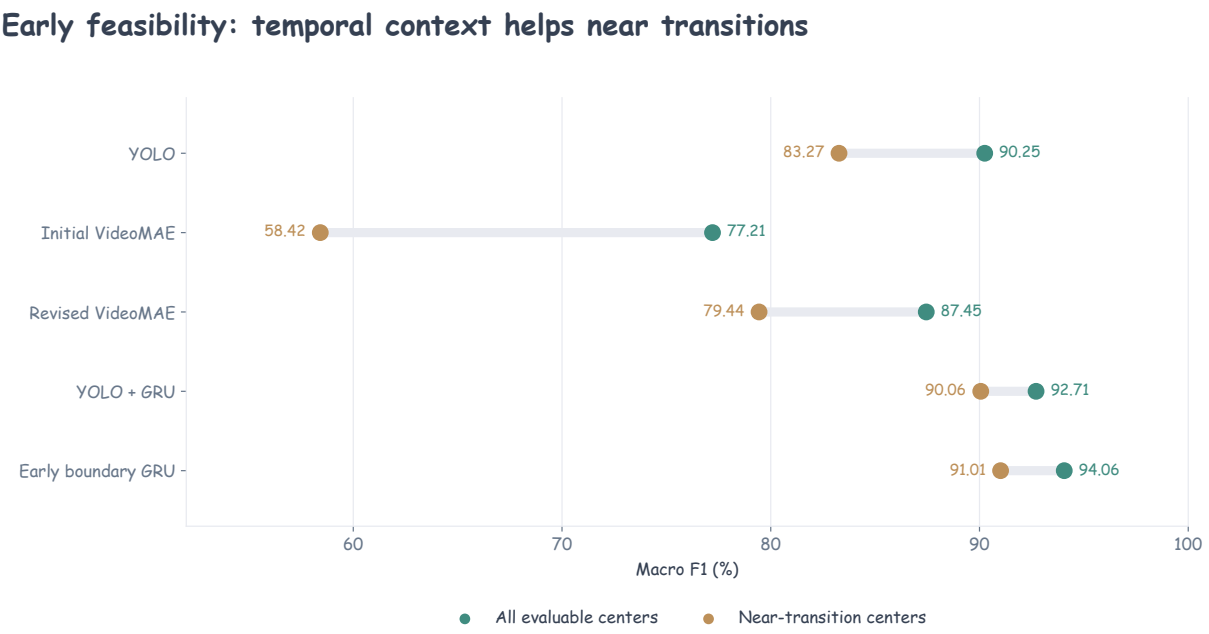


Figure 4: Archived PSKUS feasibility results. Connected points compare overall and near-transition macro F1 within each method; this public-data experiment uses a different protocol from the final target-scene evaluation.

The revised VideoMAE feasibility run also improved over its initial configuration, from 77.21% to 87.45% overall macro F1. The retained analysis attributes this change to the treatment of windows containing neighboring actions while retaining a stable center target. This experience shows why a video classifier's sampling and labeling contract must be examined alongside its architecture. The name of a model alone does not determine the quality of the experiment.

2.4 Lessons carried into target-scene development

The public-data study contributed three practical lessons. Temporal context was useful; near-transition behavior required explicit analysis; and background prevalence could distort the interpretation of accuracy. Other accounted for 69.7% of valid centers in that preliminary protocol, making class-balanced reporting particularly useful. The strict adjacent-change boundary-head score from the early study measured a different object from the output-boundary metric used later, so the two are not treated as directly comparable.

The final models were developed on the self-recorded scene. They use neither PSKUS training samples nor PSKUS-trained weights. Generic VideoMAE pretraining remains part of its initialization, and is identified separately. This progression treats PSKUS as feasibility evidence and the self-recorded corpus as the source of target-scene fitting and evaluation. It supports a coherent technical narrative without implying that the final model was trained on all data mentioned anywhere in the project.

3 Target-Scene Data and Annotation

3.1 Corpus composition and video-level partitions

The self-recorded corpus supplies 6,987 sampled frames with annotation records across 32 videos [7]. It contains continuous standard procedures, deliberately nonstandard orders, omissions and isolated recordings of Steps 4 and 5. Continuous recordings are needed to evaluate transitions and complete action chains. Isolated gestures provide concentrated visual examples, but they do not create evidence about ordering a complete procedure. The two recording types therefore have complementary roles rather than interchangeable evaluation meanings.

The original development partition contains 14 training videos and eight validation videos; ten videos form the reported evaluation set. Figure 5 shows both the number of videos and the number of sampled frames in each partition. The distinction matters because a larger number of neighboring frames increases coverage of appearances and temporal variation without creating an equivalent number of independent recordings.

Table 2: Video-level partitions and frame annotation counts

Partition	Videos	Sampled frames	Valid labels	Uncertain frames
Train	14	2901	2822	79
Validation	8	1364	1333	31
Test	10	2722	2652	70
Total	32	6987	6807	180

Table 2 separates sampled records from valid classification targets. Of the 6,987 records, 6,807 have valid class labels and 180 mark uncertain transitions. The training, validation and test subsets contain 2,901, 1,364 and 2,722 sampled records respectively. These counts are obtained from the retained frame files rather than inferred from approximate video durations. A validity mask determines which records contribute to classification metrics and supervised class losses.

The final temporal-head fitting procedure subsequently combines the 22 development videos after model selection. That operation changes the final fitting set for the recurrent heads, but it does

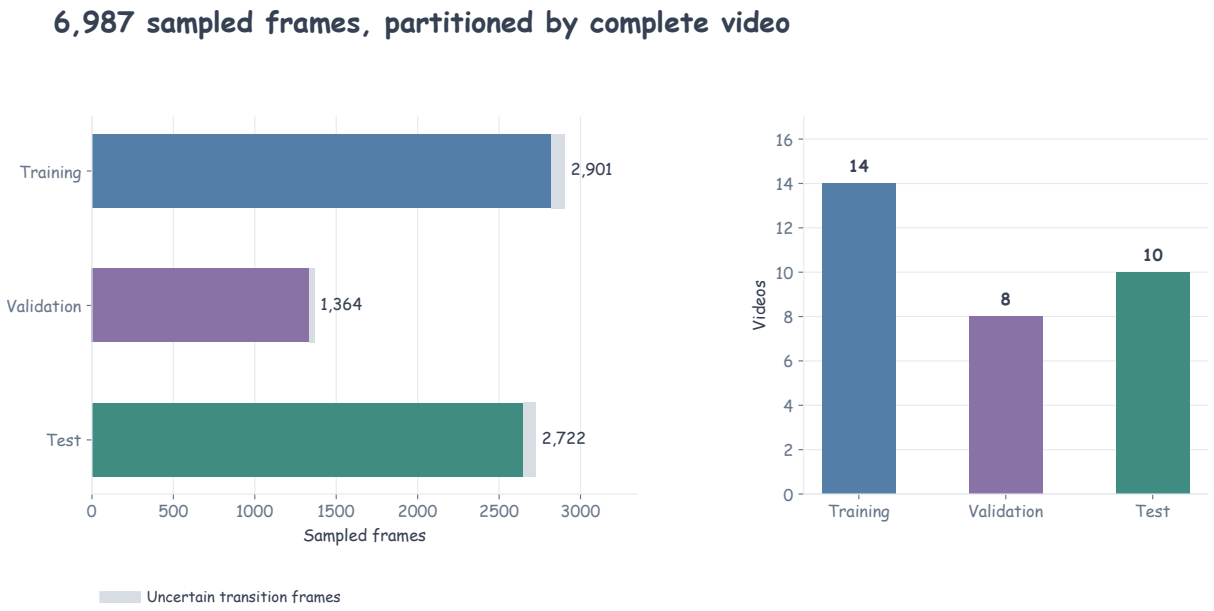


Figure 5: Video-level partitions and annotation coverage. The shaded end portions indicate uncertain transition frames; all frames from one recording remain in the same partition.

not move the ten evaluation videos into training. Section 7 explains this distinction in detail. The original partition names are retained in the dataset statistics because they describe the selection protocol and the lineage of the records.

3.2 Class coverage and recording duration

Figure 6 breaks valid frame counts down by class and partition. Dorsum rubbing and backs-of-fingers rubbing have substantial representation, while the isolated Step 4 and Step 5 recordings contribute targeted coverage of those gestures. The figure is useful for identifying how the sampled corpus is distributed; it is not a claim that exposure to every class is equal or that a high frame count removes viewpoint dependence.

Frame counts should be interpreted together with recording coverage. Figure 7 displays the sampled time span of each clip and distinguishes continuous recordings from isolated gestures. This view exposes differences that a total frame count would conceal: some recordings contain complete multi-action sequences, whereas others concentrate on one gesture. The plotted quantity is the last sampled timestamp minus the first. It is not the container's exact encoded media duration, and it does not include unsampled portions before or after those timestamps.

The target sampling rate is approximately 5 fps. Actual timestamps are retained because decoding and acquisition do not necessarily produce an exactly uniform sequence. During retrospective comparison, timestamps identify the same observation across model outputs. During application assessment, timestamps delimit the amount of observed time. A nominal frame rate is useful for specifying the intended stream, but it should not replace timing information when actual observations are available.

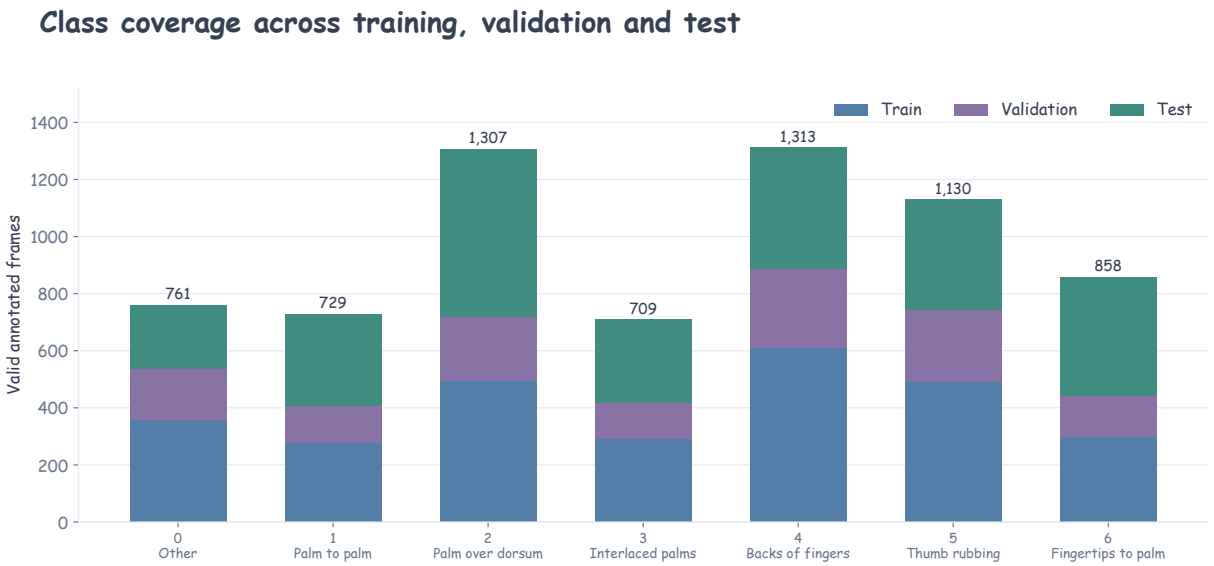


Figure 6: Valid annotated frames per class, stacked by partition. Stacked counts distinguish the partitions within each action class.

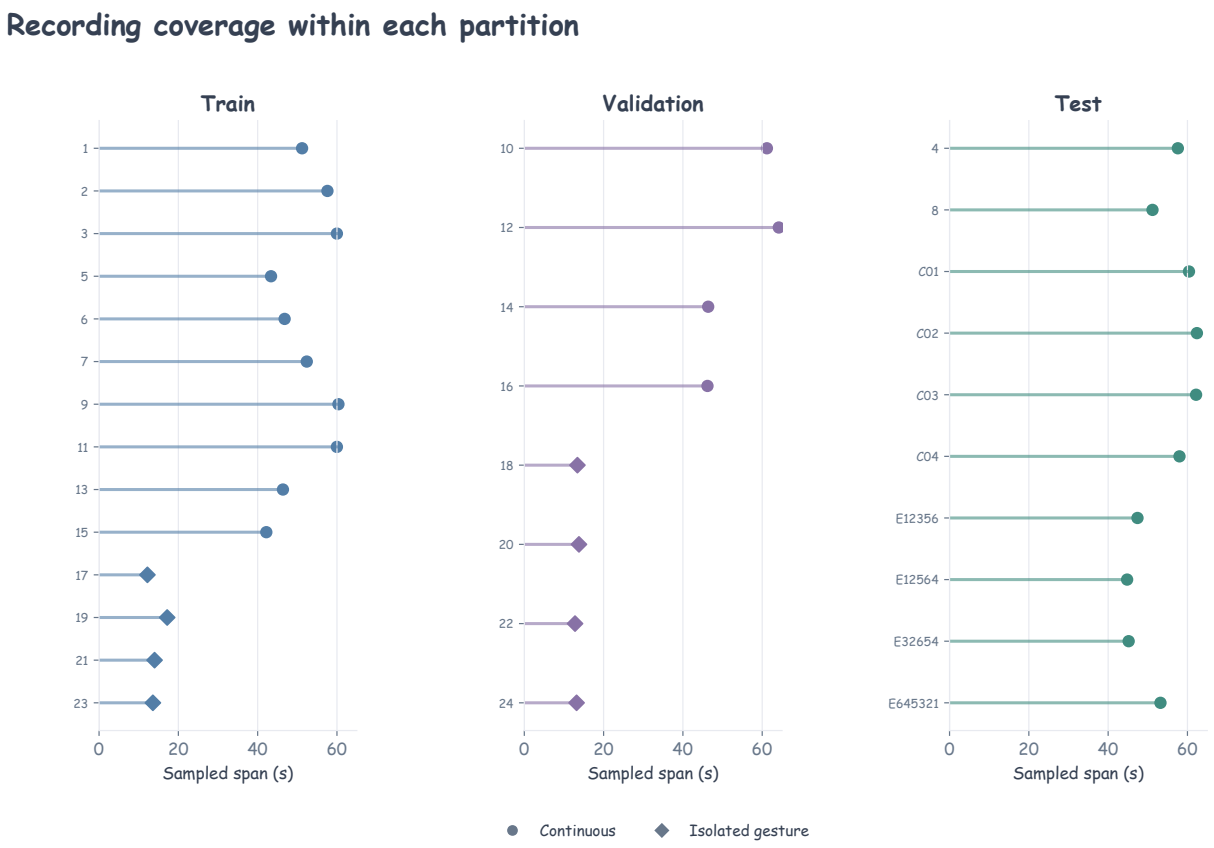


Figure 7: Sampled time span for each recording, grouped by partition. Circles mark continuous recordings and diamonds isolated gestures. C and E abbreviate supplemental standard and nonstandard test recordings.

3.3 Interval annotation and frame targets

The annotation process begins with time intervals describing the visible action. These intervals are mapped to sampled frame records, producing one training or evaluation target per sampled image. This approach retains the efficiency of interval annotation while yielding the frame-level supervision needed by the recognition pipeline. The image identity, sampling time, class ID and validity flag form the core of each record.

Figure 8 shows the final annotation for original Video 1. The long action intervals preserve within-action variation, including movement of the wrists and exchange of the active hand. Short uncertain gaps are kept separate where an unambiguous gesture cannot be assigned. This makes the distinction between a stable action interval and a transition visible in the training data rather than leaving it to an implicit convention.

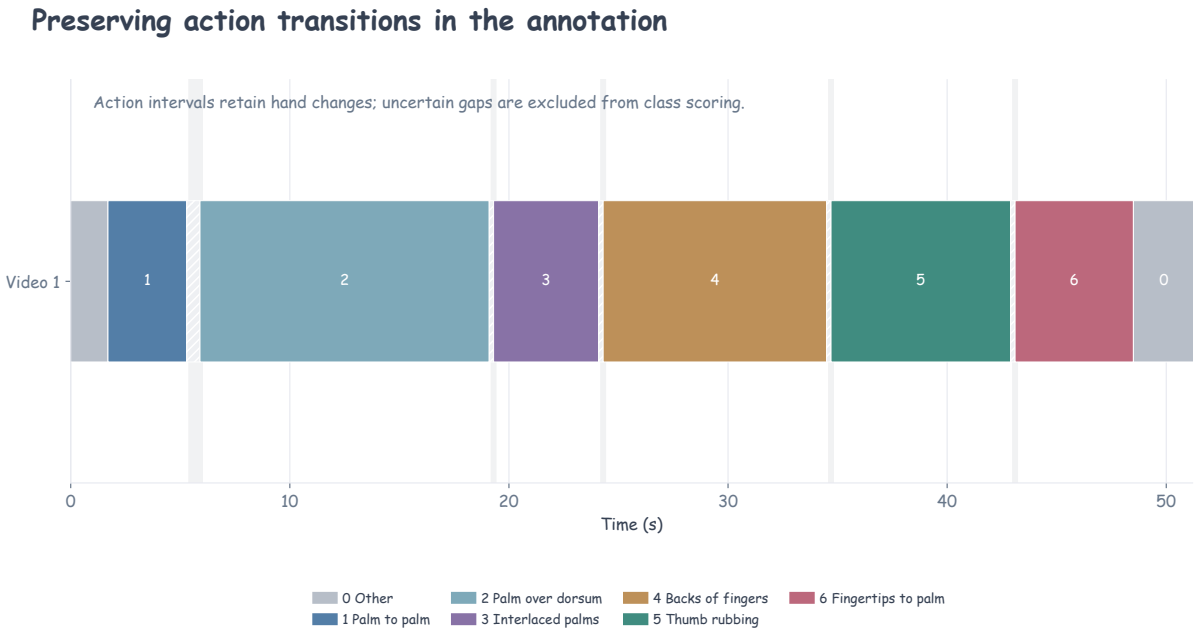


Figure 8: Frame-level labels derived from the final intervals for Video 1. Hatched gaps are uncertain transitions excluded from class scoring, rather than an additional predicted class.

There is a trade-off at action boundaries. Assigning every transitional frame to a neighboring action produces apparently complete labels, but may create false precision about the exact moment of change. Marking an excessively broad region uncertain removes useful supervision. The retained records make the uncertainty explicit and let the boundary metric operate on transition intervals. This avoids demanding that a model recover a unique change instant where the annotation itself only supports a range.

Other serves a different purpose from uncertainty. It denotes a valid observation of background or non-target activity. A frame can be confidently Other, just as it can be confidently Step 1. An uncertain transition instead indicates that a class target should not be used for scoring. Treating the two as the same state would teach the model that ambiguous movements necessarily look like background and would alter the denominator of the classification metrics.

3.4 Label review and the Step 4/5 correction

Steps 4 and 5 require careful review because hand enclosure can make their appearances superficially similar. The relevant distinction is the action being performed: backs of fingers against the opposing palm versus rotational rubbing of an enclosed thumb. A review therefore needs image evidence across a short interval, not just the recording filename or the presumed standard order. Class IDs, displayed names and frame timestamps must be checked at the same time.

The final reference for original Video 8 is 124536. Step 4 occupies the reviewed interval through 28.6 seconds, a transition is left uncertain to 29.0 seconds, and Step 5 starts at 29.0 seconds. All final results in this report use the corrected reference. Earlier exports with a different interpretation are historical records, not alternative ground truths to average together. The dedicated case study later in the report relates this correction to dense prediction and confidence traces.

Because the evaluation videos were repeatedly inspected, including this correction after predictions had been viewed, their role is retrospective internal validation. This does not make the annotation correction inappropriate; it makes the evaluation history relevant to the strength of the conclusion. The retained images, records and manifests allow the final reference to be inspected independently. A future independent evaluation should freeze annotation and model-selection rules before its predictions are examined.

4 Visual and Temporal Representation

4.1 A common image-feature path

The YOLO family uses a YOLO26m classification network. In this project, YOLO is a frame classifier and visual feature extractor; the reported recognition results do not establish hand-box detection or tracking performance. The checkpoint's inference size is 1088 by 1088 pixels. The camera path first fits the image into a 1080-pixel square using replicate padding and then applies the checkpoint preprocessing. Keeping these two sizes distinct avoids describing the requested preprocessing size as the model's actual supported tensor size.

The visual representation is a 1,280-dimensional feature vector. It is used directly by the image classification path and as the input to the temporal heads. Let I_t denote the sampled image and F the frozen visual feature extractor. The temporal representation begins with

$$\mathbf{f}_t = F(I_t) \tag{1}$$

$$\widehat{\mathbf{f}}_t = (\mathbf{f}_t - \boldsymbol{\mu}) \oslash \boldsymbol{\sigma} \tag{2}$$

$$\mathbf{e}_t = \phi(\widehat{\mathbf{f}}_t) \tag{3}$$

Here $\oslash$ denotes element-wise division. The stored normalization statistics belong to the corresponding final fitting procedure; they must travel with the temporal checkpoint. The projection ϕ includes a linear mapping to 128 dimensions, a GELU nonlinearity and dropout in the training path. The retained dropout setting is 0.3.

Freezing the extractor reduces the temporal training problem to learning a model over a fixed feature sequence. It also gives the ordinary and boundary-aware recurrent heads the same type of visual input. This supports a meaningful comparison within the YOLO family, while the later

differences in fitting and decoding still need to be reported. A shared feature source does not by itself turn the full experiment into a controlled module ablation.

4.2 Unidirectional recurrent context

The GRU updates a hidden state using the current projected feature and the preceding state. Its class probabilities are obtained through a linear head and softmax:

$$\mathbf{h}_t = \mathrm{GRU}(\mathbf{e}_t, \mathbf{h}_{t-1}) \tag{4}$$

$$\mathbf{p}_t = \mathrm{softmax}(W_c \mathbf{h}_t + \mathbf{b}_c) \tag{5}$$

The hidden dimension is 128. The seven coordinates of $\mathbf{p}_t$ correspond to Other and the six target actions. The ordinary GRU baseline uses this recurrent classification path without the proposed adaptive confirmation stage.

A unidirectional state is useful in the camera application because it can be updated as observations arrive. However, this statement applies to the recurrent backbone, not automatically to every operation in the final system. Centered smoothing and recording-level segment statistics can use observations beyond an earlier instant. Live labels are therefore provisional, while the finalized chain reflects the evidence available after the session. Distinguishing backbone causality from final-decoder causality is necessary when describing real-time behavior.

The hidden state should also be associated with a single session. Carrying it from a previous recording would mix temporal evidence from unrelated procedures. Model selection is locked during acquisition, and a new session requires a new temporal state. The same principle applies to counters, cached predictions and duration accounting: their lifecycle must follow the current recording rather than the lifetime of the web page.

4.3 Different temporal input contracts

Figure 9 compares the information available to the three representation types. The image classifier uses the current frame. The recurrent model uses the current feature and its accumulated state. VideoMAE processes a 16-frame window. At the nominal 5 fps sampling rate, the span between the first and last samples in that window is three seconds. Under the retained centered-target convention, seven samples lie after the target, giving approximately 1.4 seconds of future context.

This diagram explains why equal scoring timestamps do not imply identical information access. All models can be evaluated against the same center label, yet a centered video window may see future images that an immediate frame update has not received. That is a legitimate difference between configured systems, provided it is explicit. A camera-oriented decision must consider both recognition performance and when the supporting evidence becomes available.

VideoMAE uses 224 by 224 images inside each window, rather than the YOLO path's 1088-pixel input. The lower spatial resolution does not make it an inferior reference: the video representation, pretraining and temporal context are also different. Its observed accuracy in the retained experiment demonstrates that resolution alone does not explain the ranking. Conversely, the results do not establish that a different resolution would leave the ranking unchanged, because no matched resolution study is reported.

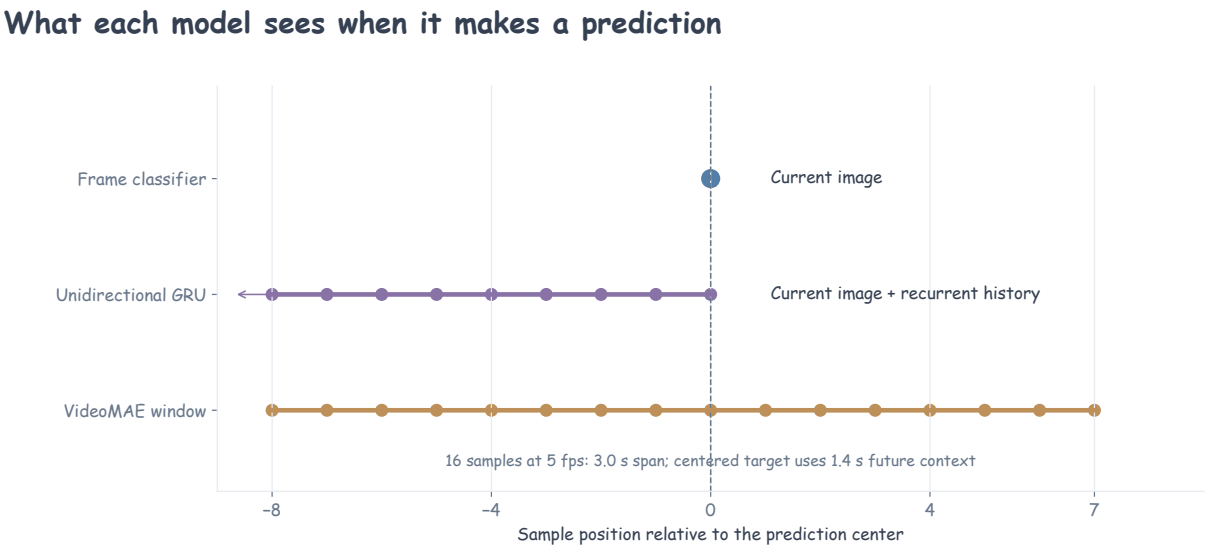


Figure 9: Input-context contracts for frame classification, recurrent encoding and the retained VideoMAE window. The recurrent backbone is unidirectional; final segment confirmation can use later session evidence.

4.4 Representation contracts in the delivered system

The practical model interface must specify more than a filename. It needs an ordered class mapping, expected input geometry, feature dimension, normalization statistics and any temporal state. A boundary-aware checkpoint additionally needs its decoder parameters. VideoMAE requires its model configuration and image preprocessing configuration alongside the weights. These are part of the model contract, not optional documentation details.

The report uses the word confidence for a model probability associated with a predicted class. It is not interchangeable with measured accuracy. For a new unannotated camera recording, the interface can display confidence but cannot compute an empirical accuracy score. A high-confidence confusion can still occur when camera placement or hand appearance differs from the learned scene. That observation motivates the detailed error analysis rather than assuming that confidence verifies the class mapping or the gesture itself.

5 Boundary-Aware Temporal Modeling

5.1 Why boundaries need their own representation

The ordinary recurrent classifier predicts a class at every sampled instant. A change of its most probable class creates a candidate action boundary, but that change can result from either a genuine transition or a transient classification error. Boundary-aware modeling introduces a learned signal related to change, allowing the temporal structure to be represented explicitly rather than inferred solely from an unregularized sequence of class winners.

The boundary-aware GRU is a neural model with a shared recurrent representation and separate class and boundary outputs. Boundary awareness is therefore not a completely independent algorithm substituted for the GRU. The recurrent state remains the feature on which both branches operate. The final named pipeline additionally includes boundary-based decoding and adaptive segment confirmation. Keeping the neural branch and the downstream decision stage conceptually separate makes it possible to explain their roles without claiming that they are unrelated systems.

For the retained architecture, the boundary branch receives the current hidden state and the magnitude of its change:

$$\Delta \mathbf{h}_t = \mathbf{h}_t - \mathbf{h}_{t-1} \tag{6}$$

$$q_t = \text{sigmoid}(g_\theta([\mathbf{h}_t; |\Delta \mathbf{h}_t|])) \tag{7}$$

The brackets denote concatenation. The boundary multilayer perceptron maps 256 input coordinates to 64 hidden units and then to one output. This construction makes both the current state and the amount of state change available to the branch. It does not prove that every large hidden-state change corresponds to an action boundary; that relationship is learned under the supplied supervision.

5.2 Classification and interval supervision

The archived final boundary-head training objective combines class cross-entropy, direct binary boundary supervision and an interval term. With the retained coefficients, it is

$$\mathcal{L} = \mathcal{L}_{\text{CE}} + 0.35\mathcal{L}_{\text{BCE}} + 0.5\mathcal{L}_{\text{int}} \tag{8}$$

These coefficients describe the delivered training source. They are not inferred from the final results, and they should not be replaced in the report by values from an earlier proposed experiment. The checkpoint and the script that produced it define the method being evaluated.

Let $\mathcal{I}$ be a collection of annotated transition intervals, each of which contains a change. The interval term uses a noisy-OR form:

$$\mathcal{L}_{\text{int}} = -\frac{1}{|\mathcal{I}|} \sum_{J \in \mathcal{I}} \log \left(1 - \prod_{t \in J} (1 - q_t) \right) \tag{9}$$

The product is small when at least one sampled point in the interval has strong boundary evidence. The resulting objective encourages evidence somewhere within the permissible interval without requiring every point in it to be a boundary. This matches the annotation's temporal precision more closely than pretending that all uncertain transitions have a uniquely known change frame.

The class objective and the boundary objective serve different purposes. Classification identifies what is happening, while the boundary terms address where an action interval changes. Improving one does not guarantee improving the other. A model could produce smooth class sequences while missing genuine transitions, or retain sharp changes while confusing neighboring classes. The result section therefore reports both class metrics and output-boundary behavior, rather than treating the presence of a boundary branch as evidence that segmentation must improve.

5.3 From boundary evidence to candidate segments

The retained decoder selects local boundary peaks using a threshold of 0.3 and a minimum gap of four sampled points. At the nominal sampling rate, that gap corresponds to 0.8 seconds, but its implemented unit is a sample count. The selected peaks partition the sequence into candidate segments. Class probabilities are averaged within a segment, and the resulting segment label supplies the decoded class sequence used by the next stage.

Segment averaging is useful when small fluctuations occur within an otherwise coherent action. It can also overwrite valid local class evidence if a boundary is missed. For example, a short Other

interval at the start of an action may be absorbed into the dominant segment class. This is a real design trade-off: the same operation that suppresses isolated errors can reduce sensitivity to short genuine states. The report consequently retains Other-class performance and boundary misses instead of evaluating only the six action classes that are easiest to describe.

No normal-order transition graph is imposed. The decoder can represent a change from Step 6 to Step 4 just as it can represent Step 1 to Step 2. This is necessary for the deliberately nonstandard recordings. A boundary detector should find the observed division, not infer the division most compatible with a prescribed procedure.

5.4 Boundary evidence versus the reported boundary metric

Figure 10 explains the interval-based evaluation. Reference transitions are represented by the interval between the last valid sample of one label and the first valid sample of the next. Predicted events are derived from changes in the class output, placed between consecutive prediction times. The metric therefore assesses the final label sequence's change structure, not the calibration of the internal q_t values.

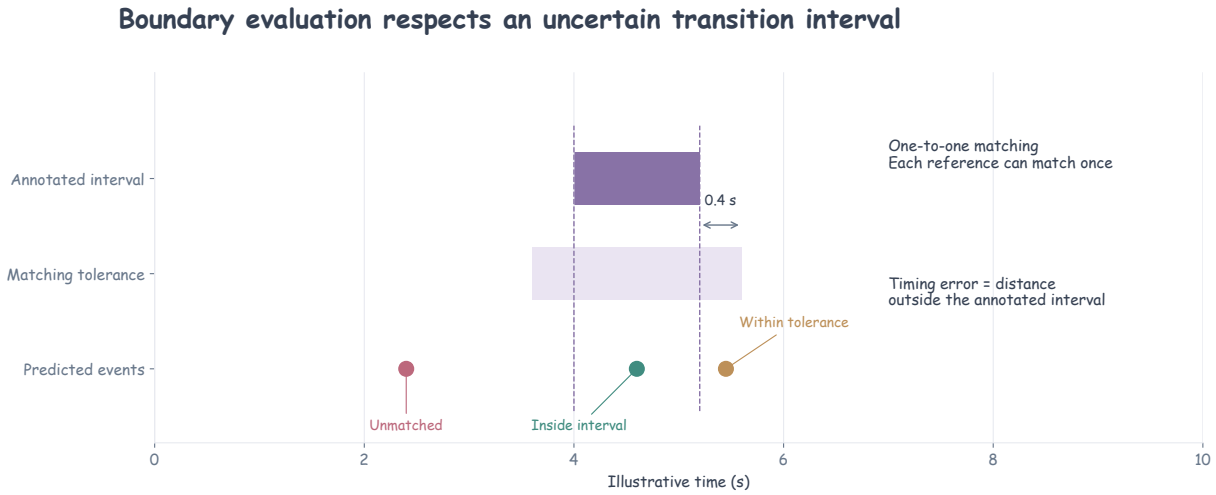


Figure 10: One-to-one matching of predicted label changes to annotated transition intervals, expanded by 0.4 seconds. The positions are schematic; only one candidate can match a particular reference transition.

The distinction prevents two common misinterpretations. First, a high output-boundary F1 does not directly establish that the boundary probability head is well calibrated. Second, a zero timing error does not imply that the exact transition instant is known. An event anywhere inside the reference interval has zero interval distance. Section 8 defines the pooled metric formally, and the results show detection counts alongside the conditional timing error.

This design also makes missed and extra boundaries visible. Increasing tolerance can make an event easier to match, but one-to-one matching still prevents several nearby predictions from receiving credit for the same reference. Extra events remain false positives. The resulting measure is particularly relevant to an application where over-segmentation can create spurious action-chain steps.

6 Adaptive Action Confirmation

6.1 Replacing a fixed admission duration

A fixed minimum segment duration is easy to implement, but it conflates two different questions: whether an interval contains a recognizable action, and whether that action is long enough relative to the current recording to be a credible separate step. People and recordings can differ in pace. A duration that is typical in one sequence can be unusually short in another. A rigid three-second rule also gives no role to confidence, even when a short segment is visually well supported.

The proposed confirmation stage uses recording-relative statistics instead. It first smooths labels over approximately one second and groups adjacent equal labels into segments. For each non-background segment, it calculates duration and class-confidence evidence. A candidate is left unconfirmed only when it is both an unusually short duration outlier and weaker than the typical segment confidence in that recording. Figure 11 summarizes the computation.

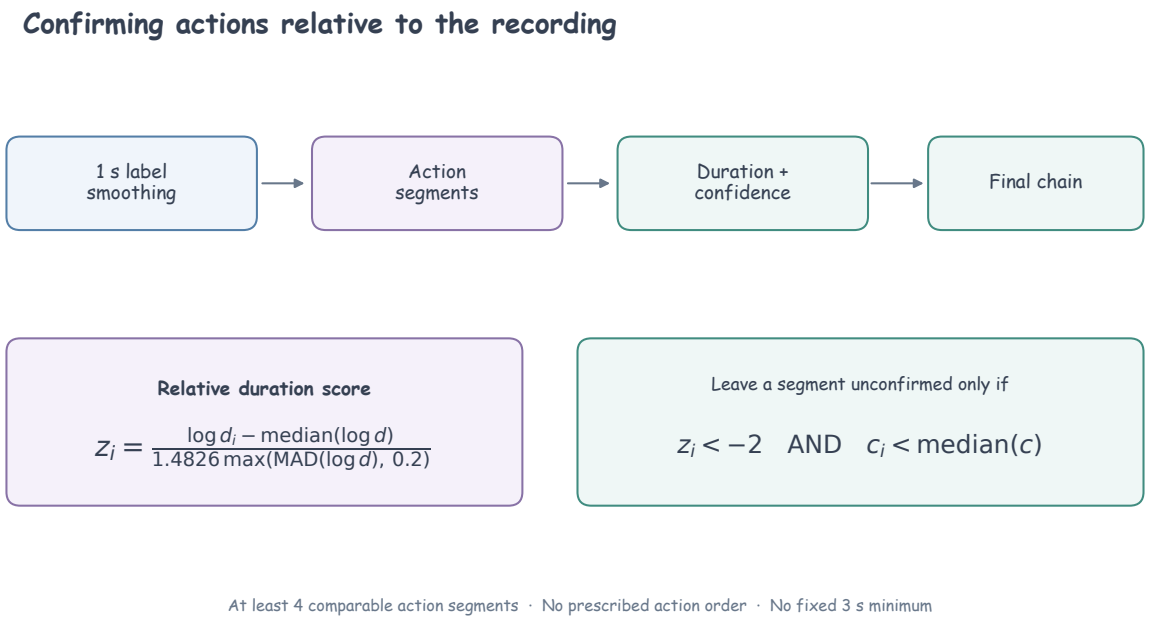


Figure 11: The implemented adaptive confirmation rule. Relative log-duration and confidence jointly determine whether a candidate remains unconfirmed; a normal action order is not supplied to the decision.

The procedure remains parameterized. Removing a fixed three-second threshold does not remove all thresholds or make the algorithm universally pace invariant. The one-second smoothing window, the robust-scale floor and the outlier cutoff still define its behavior. The benefit is a more explicit use of within-recording evidence, with a rule that can be inspected for each rejected segment.

6.2 Smoothing and segment construction

At the nominal 5 fps rate, the one-second smoothing window corresponds to five samples. The implementation converts the requested duration to an odd window length, so that a central sample is well defined. It applies majority voting in the local neighborhood. If several labels tie and the current label is among them, that label is retained; otherwise a tied label encountered in the

neighborhood is selected. The tie rule matters for reproducibility because ties can affect short runs at transitions.

After smoothing, consecutive equal labels form segments S_i. Their sampled duration is $d_i = |S_i|/f_s$, where f_s is the nominal sample rate used by this retrospective decoder. Background segments remain in the temporal record but are excluded from the reference population used to assess action duration. The action chain is constructed only after confirmation decisions are made.

When complete class probabilities are available, the confidence of a segment is the mean probability assigned to that segment's class. Historical exports sometimes retain only the original winning label and its confidence. For those exports, the audit averages winning confidence only at samples whose original winning class agrees with the smoothed segment class. It does not reconstruct a full probability vector from a single stored number. This distinction is important when interpreting the later confidence scatter plot.

6.3 Robust duration and confidence statistics

Let $\mathcal{A}$ index the non-background segments. Duration is placed on a logarithmic scale, and a robust location and scale are computed as

$$m = \operatorname{median}_{i \in \mathcal{A}} (\log d_i) \tag{10}$$

$$s = 1.4826 \max \left\{ \operatorname{median}_{i \in \mathcal{A}} \left| \log d_i - m \right|, 0.2 \right\} \tag{11}$$

$$z_i = \frac{\log d_i - m}{s} \tag{12}$$

$$\widetilde{c} = \operatorname{median}_{i \in \mathcal{A}} c_i \tag{13}$$

The log transform expresses duration differences relatively rather than as a fixed number of seconds. The median limits the influence of a very long segment on the location estimate. The scale floor prevents nearly identical durations from creating an arbitrarily small denominator. These choices explain the implementation; the retained experiment does not establish that they are individually optimal.

With at least four comparable action segments and available confidence, the rejection condition is

$$\text{unconfirmed}(S_i) \iff z_i < -2 \ \wedge \ c_i < \widetilde{c} \tag{14}$$

The conjunction is essential. Short duration alone is insufficient when the segment has strong confidence relative to its peers. Low confidence alone is insufficient when the interval is not a marked short-duration outlier. This preserves a distinction between an uncertain interval and an implausibly brief candidate step.

When fewer than four action segments are available, duration-outlier rejection is skipped. The within-video reference distribution is then too sparse for the retained comparison rule. If historical confidence is missing, the implementation treats it as below typical confidence; the deployed interface supplies probabilities. These edge conditions are included in the algorithm description so that the displayed formula is not mistaken for the entire implementation contract.

6.4 Chain formation and its limits

Confirmed non-background segment labels are collected in their observed temporal order. Adjacent equal confirmed labels are collapsed, yielding the final action chain $\mathcal{C}$. The standard sequence is not consulted during this construction. As a result, the algorithm can retain a sequence such as 645321 or a sequence with a missing step. A faithful nonstandard chain should be returned to the assessment layer, which can then explain why the procedure is incomplete.

Algorithm 1 Adaptive action-chain confirmation

Require: Decoded labels $y_{1:T}$, probabilities or archived winner confidence, sample rate f_s
Apply the odd one-second majority window and construct equal-label runs S_i
Record each run's class, sample extent, duration d_i and available confidence c_i
Mark background runs as background and action runs as confirmed
Let $\mathcal{A}$ contain the non-background runs
if $|\mathcal{A}| \geq 4$ **then**
    Calculate m, s, z_i and median available confidence
    for each $i \in \mathcal{A}$ **do**
        if $z_i < -2$ and (confidence is below the median or unavailable) **then**
            Mark S_i unconfirmed
        end if
    end for
end if
Collect confirmed action classes in temporal order
Collapse adjacent equal classes and return the chain with all run statuses

Leaving a segment unconfirmed does not erase it from the evidence record. Its position, duration, confidence and status remain available for inspection. It also does not necessarily remove its time from recognized washing duration, because the duration assessment answers whether a washing action was observed, not whether that interval should become a distinct chain step. This separation is particularly relevant for short interruptions within an otherwise continuous washing procedure.

The method cannot guarantee that every rejected segment is false. A genuinely brief, low-confidence action may satisfy the rejection condition. Conversely, a high-confidence error may survive. The observed benefit must therefore be established from the retained videos and inspected segment decisions. Section 10 examines the actual 1.8-second and 2.8-second candidate runs that explain two final-chain corrections, while the discussion identifies independent short-action recordings as a useful future evaluation target.

7 Training Protocol and Model Provenance

7.1 Development partition and final fitting

The retained training workflow distinguishes selection from final fitting. YOLO and VideoMAE use the 14-video training partition and eight-video validation partition. The recurrent heads are first selected on that same development split, but the archived script subsequently reinitializes and fits them on all 22 development videos. Their final normalization statistics are also computed from those 22 videos. The ten evaluation clips are absent from this final-fit list.

Figure 12 makes this sequence explicit. It is included before the results because the final fitting history affects the interpretation of the comparison. A result obtained by a refitted temporal head

cannot be described as though it came from the earlier selected checkpoint with unchanged normalization. Likewise, a claim of equal training quantity across all four models would be inaccurate. The reported comparison concerns the four delivered pipelines with their actual provenance.

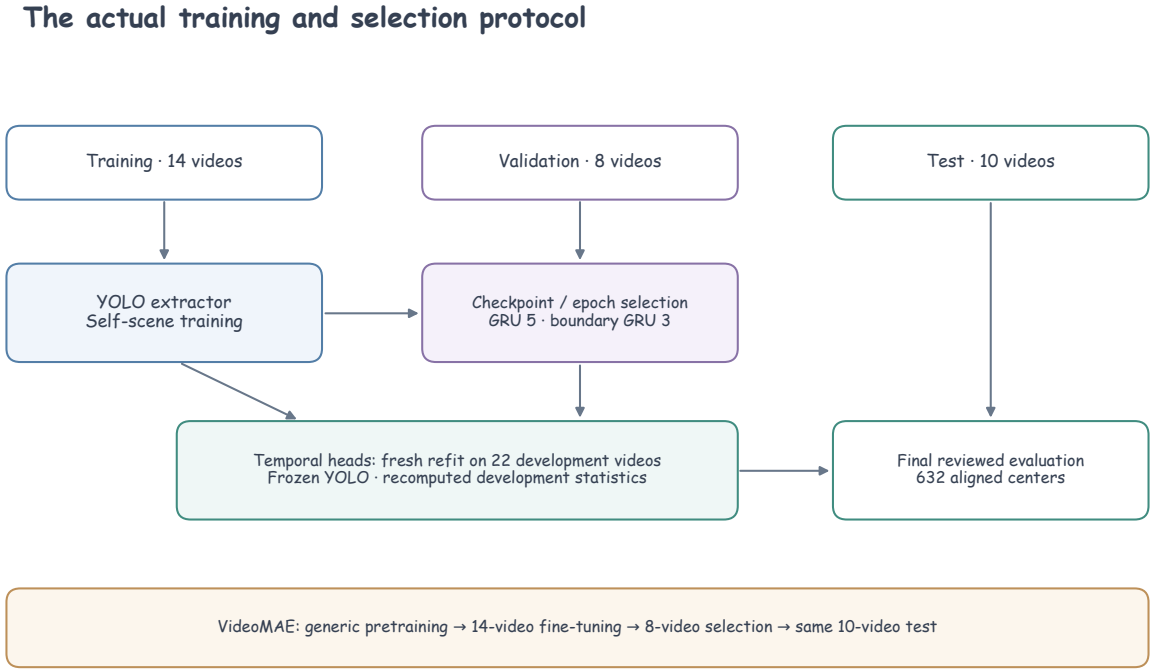


Figure 12: The retained selection and fitting procedure. Temporal heads are selected on the 14/8 development split and then freshly refit on all 22 development recordings; VideoMAE retains its 14/8 fine-tuning and selection procedure.

7.2 Visual and recurrent optimization

The YOLO training source constructs the classification architecture from its YAML definition with pretrained weights disabled. Thus, its target-scene visual model is trained from random initialization. This differs from VideoMAE's generic pretraining, and the distinction is recorded in Table 3. Referring to all four systems simply as models trained on the same dataset would conceal this difference in prior representation learning.

The recurrent heads consume frozen visual features. Their archived optimization uses AdamW with learning rate 0.001 and weight decay 0.0001, together with gradient clipping. The selection procedure has a maximum of 50 epochs and a patience of ten epochs. The retained epoch counts are five for ordinary GRU and three for the boundary-aware model. These counts govern the later fresh fit; they do not mean that both models converge at the same rate or have been compared under every possible training budget.

The final model configuration binds together the learned parameters, normalization vectors, class order, input feature dimension and decoder settings. A recurrent weight file combined with normalization statistics from another fit is a different system, even if its tensor shapes load successfully. The delivered manifests therefore identify complete model assets. The final saved predictions and reviewed annotations, rather than historical metrics inside a checkpoint, define the numbers presented in this report.

Table 3: Input and final fitting contracts for the four pipelines

Model	Input	Initialization	Final fitting / selection
YOLO	1088 × 1088 frame	Random initialization	14 train / 8 validation
YOLO + GRU	1280-d YOLO features	Random temporal head	Select on 14 / 8; refit head on 22
Proposed	1280-d YOLO features	Random temporal heads	Select on 14 / 8; refit heads on 22
VideoMAE	16 × 224 × 224 window	Generic Kinetics pretraining	14 train / 8 validation

7.3 VideoMAE fine-tuning

VideoMAE starts from the generic checkpoint `MCG-NJU/videomae-base-finetuned-kinetics`. It uses 16-frame windows sampled at 5 fps, 224-pixel square image preprocessing and a stride of four samples. The final target-scene fitting uses the same self-recorded data source as the other systems, with the 14/8 training-validation division. No PSKUS samples or PSKUS-trained weights are incorporated into this final model set.

This configuration allows a useful practical comparison even though it is not a strict architecture-only experiment. The image-based family uses a high-resolution visual path and compact recurrent heads. VideoMAE uses a lower-resolution multi-frame representation and generic pretraining. Both are plausible ways to solve the application problem, and the retained results show how the delivered versions behave. They cannot identify how much of a difference is attributable to pretraining, resolution, context, final fitting quantity or architecture in isolation.

A separate controlled study would need to specify which of these factors is held constant and which is varied. For example, testing only the contribution of boundary supervision would require matched visual features, fitting data, optimization, class decoding and confirmation. That study is distinct from choosing the most useful delivered camera pipeline. The present report does not relabel the complete-system comparison as such an ablation.

7.4 Data integrity and repeatability

The project records clip identifiers, partitions, frame counts and checksums. Model manifests similarly record asset sizes, hashes and class contracts. These records make it possible to determine which annotations and weights belong to an evaluation. Reproducibility here means that the reported tables can be regenerated from saved predictions and the stated metric protocol, and that the delivered application can load the identified assets. It does not mean that an arbitrary retraining run must produce identical parameters.

The source snapshots for the figures retain the precise prediction tables and annotation files used in the analysis. Exact timestamp alignment is checked before confusion matrices and class metrics are calculated. This prevents accidental nearest-neighbor pairing from making unlike observations appear comparable. The procedure also verifies that the reference label at each selected center agrees across the model exports.

Training logs and scripts document where the models came from, while the final outputs document what was evaluated. Both kinds of evidence are necessary. A training configuration alone does not establish the reported performance, and a score table without source predictions does not expose sequence-level errors. The report uses the two together, with the detailed chain tables preserved in Appendix A.

8 Evaluation Design and Metric Definitions

8.1 Exactly aligned scoring centers

The four-model classification comparison uses 632 valid centers across ten videos. VideoMAE's window-center timestamps are paired with the corresponding dense prediction timestamps using exact six-decimal time keys and the correct video identity. The selected reference labels must agree, and a center must be valid for evaluation. This produces a common observation set rather than allowing each model to be scored on whichever frames are most convenient for its input format.

The 632 centers are a subset of the test partition's 2,722 sampled records and 2,652 valid labels. The difference reflects the aligned window-center protocol and validity filtering. It should not be interpreted as the number of all available test images or the number of independent test cases. Video-level sequence metrics use ten recordings, and their denominator remains ten regardless of the number of centers within each recording.

Common-center alignment controls the reference observations used for classification, but does not equalize the context supplied to the models. VideoMAE receives a window, while the GRU retains recurrent history. The final adaptive chain uses dense predictions, as explained below. These distinctions are part of the comparison design and remain relevant after timestamps have been aligned.

8.2 Classification accuracy and macro F1

Let C_{ij} denote the confusion-matrix count for reference class i and predicted class j, including class 0. Accuracy is

$$\text{Accuracy} = \frac{\sum_{k=0}^{6} C_{kk}}{\sum_{i=0}^{6} \sum_{j=0}^{6} C_{ij}} \tag{15}$$

For class k, the F1 score and the macro average are

$$F_{1,k} = \frac{2C_{kk}}{\sum_j C_{kj} + \sum_i C_{ik}} \tag{16}$$

$$F_{1,\text{macro}} = \frac{1}{7} \sum_{k=0}^{6} F_{1,k} \tag{17}$$

All seven classes have reference support in this comparison. Macro F1 gives each class equal influence, so a weak Other class can lower the average even when most action frames are correctly recognized. This is desirable when the application needs to avoid counting background activity as washing.

The confusion matrix provides more information than a single average. It identifies which reference classes contribute errors and whether mistakes are concentrated in a particular pair. The report therefore presents both raw cell counts and row-normalized color shading. Counts preserve the number of observations, while normalization makes classes with different support visually comparable. Per-class F1 is shown alongside reference support to make the influence of the 21 Other centers visible.

8.3 Output-boundary matching

A reference boundary is an interval $[a_j, b_j]$ between consecutive valid observations whose labels differ. A predicted event is the midpoint between consecutive scoring times where the output class changes. The signed distance of event time τ from a reference interval is

$$e(\tau, [a, b]) = \begin{cases} \tau - a, & \tau < a \\ 0, & a \le \tau \le b \\ \tau - b, & \tau > b \end{cases} \tag{18}$$

A pair is eligible when the absolute interval distance is at most 0.4 seconds. Matching is one-to-one: the retained procedure maximizes the number of matches and then favors lower timing error. Matching uses event time rather than requiring identical before-and-after class identities. Class identity is evaluated separately through the confusion matrix and chain measures.

Matched events are true positives. Unmatched predicted events are false positives, and unmatched references are false negatives. Their pooled precision, recall and F1 are

$$P_b = \frac{TP}{TP + FP}, \qquad R_b = \frac{TP}{TP + FN} \tag{19}$$

$$F_{1,b} = \frac{2TP}{2TP + FP + FN} \tag{20}$$

Timing error is averaged only over matched pairs. Detection counts must therefore accompany it: a model could have a small conditional timing error while missing transitions or generating many extra ones. The zero error assigned inside each interval also means that a millisecond-scale mean does not establish millisecond-scale annotation precision.

8.4 Raw chains and confirmed chains

The raw action chain removes background labels and collapses adjacent repeated action labels from the common-center sequence. Exact-chain matching requires equality with the complete reference sequence, including its order and omissions. If a prediction adds a single extra action, the entire video fails this measure. This strictness is useful for the intended assessment task, but it means that a high frame accuracy can coexist with a low exact-chain match rate.

The proposed final chain is produced from dense 5 fps outputs using adaptive confirmation. The other pipelines retain their archived raw chains. Consequently, the raw-to-final change for the proposed method includes both a denser temporal representation and its confirmation stage. It is not an isolated estimate of the contribution of one threshold or one neural branch. The report labels the two columns separately and discusses actual segment decisions before attributing a benefit to the complete pipeline.

For reference chains $\mathcal{C}_v^*$ and predicted chains $\widehat{\mathcal{C}}_v$, the match rate is

$$M_{\text{chain}} = \frac{1}{V} \sum_{v=1}^{V} \mathbf{1}\left[\widehat{\mathcal{C}}_v = \mathcal{C}_v^*\right] \tag{21}$$

Here $V = 10$. A match to a nonstandard chain is a correct recognition result, not a passing procedural assessment. This distinction is preserved in both the per-video matrix and the camera application.

8.5 Aggregate interpretation and evidence scope

The retained recognition summary combines macro F1, final-chain match rate and output-boundary F1:

$$S_{\text{recognition}} = 0.4F_{1,\text{macro}} + 0.3M_{\text{chain}} + 0.3F_{1,b} \tag{22}$$

All components are expressed on the same scale. This score summarizes recognition behavior and excludes model size, energy, latency and memory. It cannot support a deployment-efficiency claim by itself. VideoMAE leads this summary in the retained evaluation, while the proposed method leads the YOLO family.

The ten videos have been reviewed repeatedly during development. Their results characterize internal performance under the final annotation and configuration, not a prospective estimate on untouched recordings. The report does not use adjacent frames as independent experimental repetitions or infer statistical significance from small percentage differences. Its strongest evidence is the combination of reproducible aggregate counts, complete sequence outputs and inspectable case analyses.

9 Results and Comparative Analysis

9.1 Recognition and complete-sequence performance

Table 4 reports the final common-center metrics and chain results [7]. Figure 13 presents the same headline comparison using bars for classification and boundary scores and overlaid lines for raw and final chain counts. Reading the two together is useful: the table supplies exact reported values, while the figure shows that the largest separation within the YOLO family concerns temporal structure rather than frame accuracy.

Table 4: Common-center metrics and complete-chain results

Model	Accuracy (%)	Macro F1 (%)	Boundary F1 (%)	Raw chains	Final chains
YOLO	95.25	93.84	76.92	2/10	2/10
YOLO + GRU	96.04	93.58	87.59	6/10	6/10
Proposed	96.52	93.71	94.40	8/10	10/10
VideoMAE	99.53	99.02	97.48	10/10	10/10

Standalone YOLO reaches 95.25% accuracy, but only two of its ten raw chains match their references. Adding an ordinary GRU increases accuracy to 96.04% and chain matching to six videos. The proposed model reaches 96.52% accuracy and eight raw-chain matches before its final confirmation pipeline produces ten exact matches. These results explain why frame classification alone is an insufficient acceptance criterion for the application.

Relative to ordinary GRU, the proposed method increases accuracy by approximately 0.48 percentage points and macro F1 by 0.13 points. Its boundary F1 increases by approximately 6.81 points. The size of these changes should not be blurred: the classification gain is modest in this evaluation, while the change in extra temporal events and complete chains is more substantial. The subsequent boundary and case analyses identify the observations behind that difference.

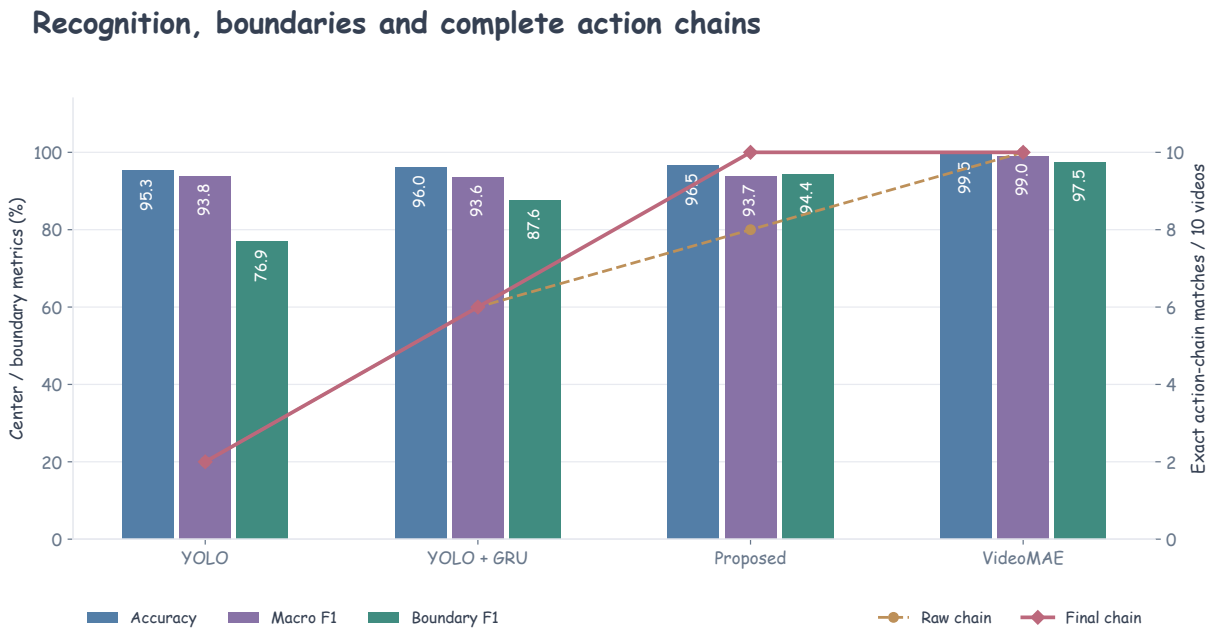


Figure 13: Four-model comparison. Bars show metrics on common centers; lines show exact action-chain matches out of ten videos. The proposed final chain uses dense predictions and adaptive confirmation.

VideoMAE achieves 99.53% accuracy, 99.02% macro F1 and 97.48% output-boundary F1, with ten exact chains. It is the strongest recognition reference in the retained comparison. The proposed method matches its final-chain result, but does not match its class metrics. The camera-model selection is therefore a multi-criterion deployment decision, not a claim that the proposed model dominates VideoMAE in predictive performance.

9.2 Confusions behind the averages

Figure 14 displays the four confusion matrices for the same 632 centers. Each cell contains a count, while the color intensity is normalized within the reference class. The diagonals show that all systems recognize much of the scene successfully. The off-diagonal cells identify the remaining weaknesses, including background handling and confusion among visually related actions.

The proposed model correctly classifies all 74 reference centers for Step 3 and all 96 for Step 5. However, perfect recall on a class does not automatically imply perfect precision: predictions from other classes can still enter that class. The per-class F1 values account for both directions of error. This is why a visual reading of the diagonal must be supplemented with numerical class metrics.

The Other class is especially informative. Its reference support is 21 centers, and its F1 is 84.00% for standalone YOLO, 70.00% for ordinary GRU, 67.92% for the proposed method and 95.24% for VideoMAE. Since macro F1 weights this class equally with each action, the proposed method's weaker Other behavior helps explain why its overall macro F1 is slightly below standalone YOLO despite its stronger temporal results. The report retains this weakness because false action time during background activity matters to the application.

Table 5 and Figure 15 also show that the historically troublesome Step 4/5 pair is not the principal source of final aggregate error under the corrected reference. The proposed method reaches 99.07% F1 for Step 4 and 100.00% for Step 5. These scores do not eliminate the need to inspect transitions,

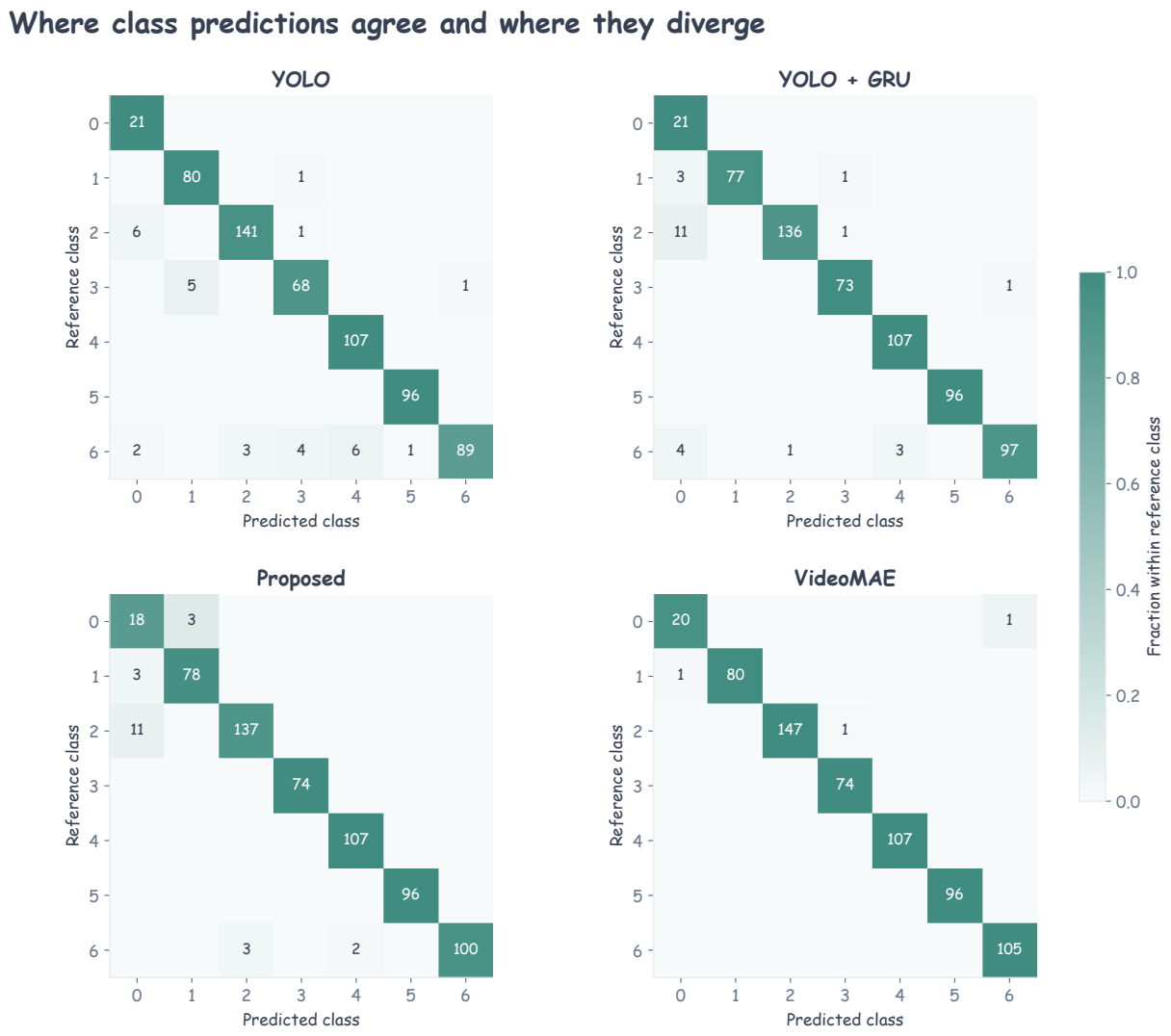


Figure 14: Confusion matrices on the common 632-center evaluation set. Cell values are counts; shading is normalized within each reference class. Class 0 is Other and classes 1–6 follow Table 1.

Table 5: Per-class F1 percentages and common-center reference support

Class	Support	YOLO	YOLO + GRU	Proposed	VideoMAE
0 Other	21	84.00	70.00	67.92	95.24
1 Palm to palm	81	96.39	97.47	96.30	99.38
2 Palm over dorsum	148	96.58	95.44	95.14	99.66
3 Interlaced palms	74	91.89	97.99	100.00	99.33
4 Backs of fingers	107	97.27	98.62	99.07	100.00
5 Thumb rubbing	96	99.48	100.00	100.00	100.00
6 Fingertips to palm	105	91.28	95.57	97.56	99.53

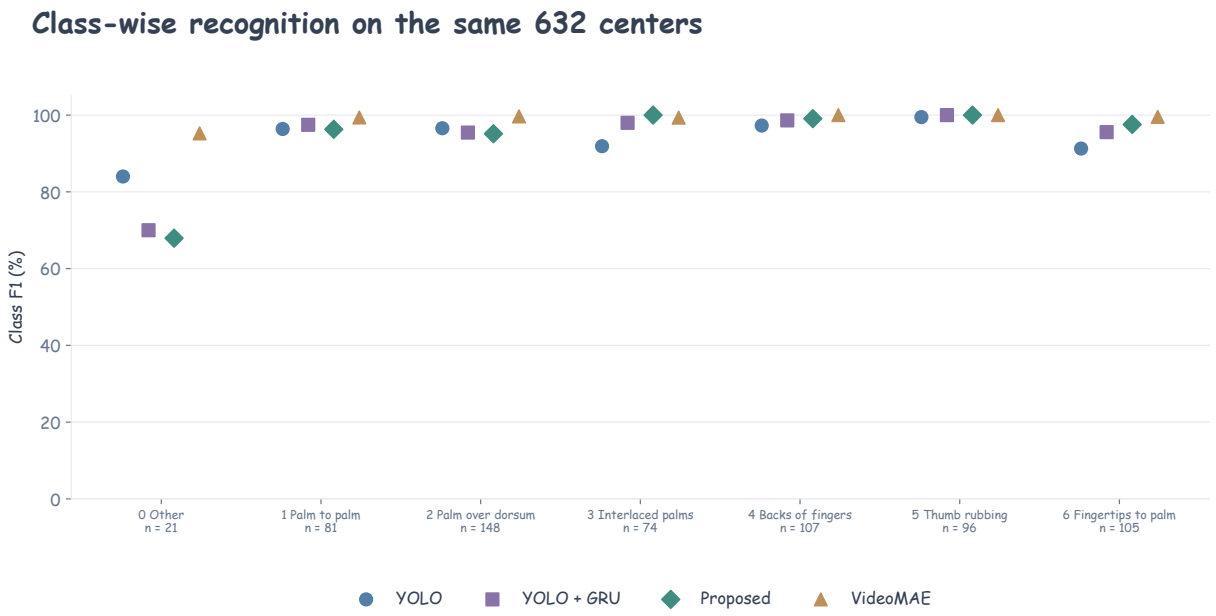


Figure 15: Class-wise F1 with reference support. The Other class has 21 common centers and materially affects the seven-class macro average.

but they redirect the final discussion from a presumed persistent class mapping failure to the remaining background and sequence issues actually present in the saved outputs.

9.3 Extra boundaries and missed transitions

The pooled boundary counts in Table 6 provide a direct explanation for the temporal improvement. YOLO produces 36 false-positive boundaries. Ordinary GRU reduces that count to 17, and the proposed method reduces it further to six. The proposed model has one missed reference transition, compared with zero for the two simpler YOLO pipelines. It improves precision substantially while accepting a small reduction in recall.

Table 6: Pooled output-boundary counts, precision, recall and matched interval error

Model	TP	FP	FN	Precision (%)	Recall (%)	Interval MAE (ms)
YOLO	60	36	0	62.50	100.00	0.00
YOLO + GRU	60	17	0	77.92	100.00	13.18
Proposed	59	6	1	90.77	98.33	13.40
VideoMAE	58	1	2	98.31	96.67	6.82

The decrease from 17 to six false positives is approximately 64.7% relative to ordinary GRU. This is a concrete change in the output sequence: fewer spurious class changes are presented as potential transitions. VideoMAE produces only one false-positive boundary and misses two references. Its boundary F1 is highest because the overall balance of matched, extra and missed events is strongest under the retained metric.

The timing panel in Figure 16 requires a different interpretation. YOLO's mean matched interval distance is zero despite its 36 extra events. This is possible because the statistic is conditional on matched events and assigns zero distance anywhere inside a reference interval. The figure therefore

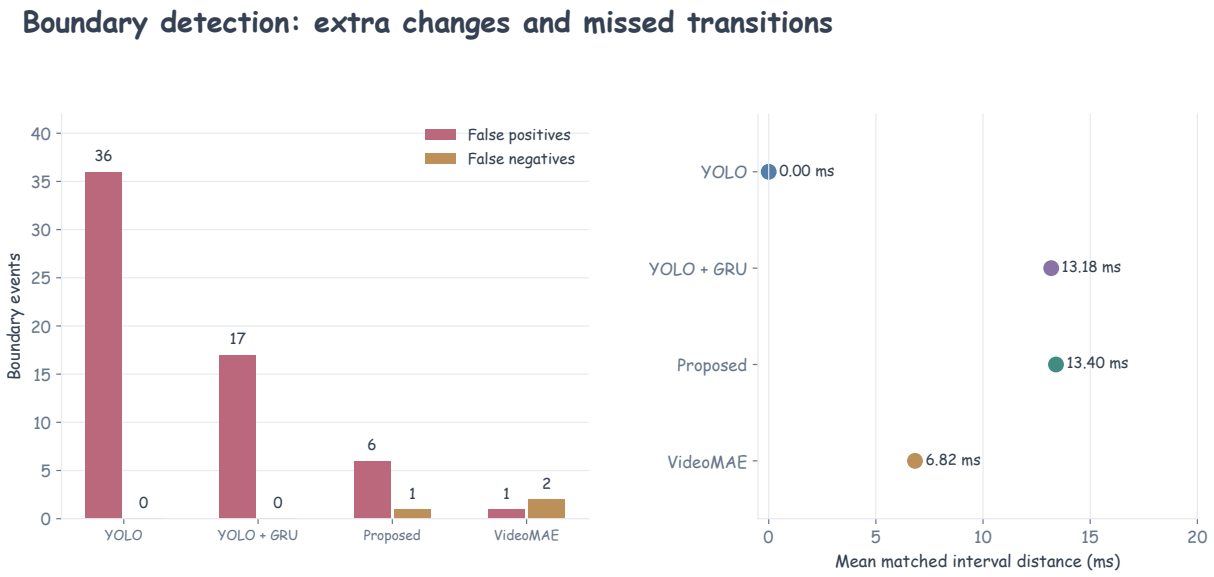


Figure 16: Extra and missed output boundaries, with conditional matched interval distance. Small timing errors must be interpreted together with event counts and the annotation intervals.

does not establish that YOLO localizes boundaries more accurately in an absolute sense. Detection quality and conditional timing are related but distinct aspects of the output.

9.4 Which complete procedures are reconstructed correctly?

Figure 17 exposes the result for every video, rather than summarizing only the number of successes. The reference chain is printed beside each recording. The proposed method and VideoMAE match all ten references. Standalone YOLO matches the nonstandard 12564 and 645321 recordings; ordinary GRU matches six videos, including several nonstandard procedures. The detailed predicted chains for all four models are listed in Appendix A.

This matrix is important for two reasons. First, it prevents an aggregate result from concealing a systematic failure on reordered procedures. Second, it separates recognition quality from procedural correctness. For example, returning 645321 for the corresponding reference is a success for the recognizer and a reason for the assessment layer to reject the standard-order criterion. A system that changed that prediction into 123456 would score poorly on faithful recognition even if its interface displayed a reassuring sequence.

The matrix also makes the strictness of exact matching apparent. An otherwise correct sequence with one extra step is a mismatch. The case studies that follow show how such failures arise and how final confirmation changes two proposed chains. They explain the mechanism behind the ten-match total without claiming that every frame or every internal boundary probability has become correct.

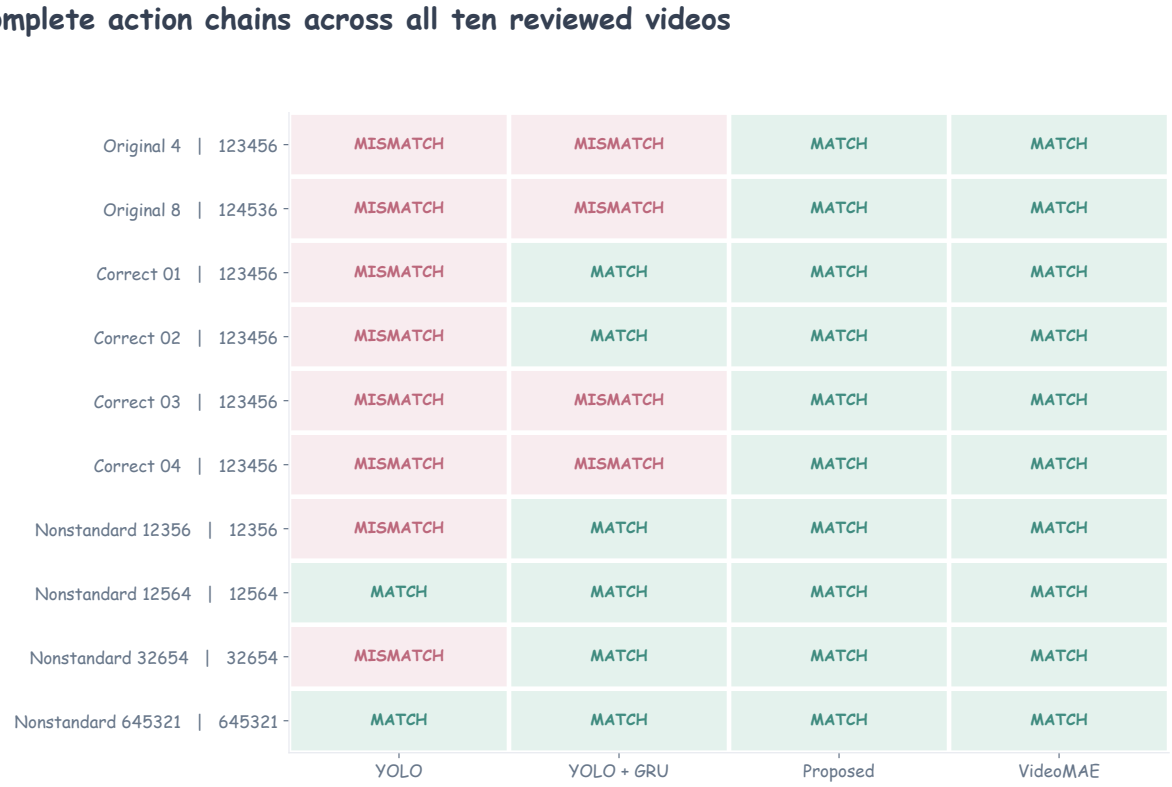


Figure 17: Final-chain agreement for all ten reviewed recordings. A match means that the observed reference order is reconstructed, even when that order is nonstandard or incomplete.

10 Temporal Case Studies

10.1 A standard procedure with an extra final step

The supplemental recording `test_correct_04` has the reference chain 123456. Figure 18 shows its class outputs at the common evaluation centers. Most of the sequence is recognized consistently across models, but the last part reveals why a whole-chain metric can fail even when the principal action intervals look correct. An extra short prediction after the final gesture introduces another class into the collapsed sequence.

The proposed raw chain is 1234562. After dense-stream smoothing and adaptive confirmation, its final chain is 123456. The ordinary GRU's final chain remains 1234562, and YOLO's includes an additional Step 3 before the final two actions. The point of the timeline is not that the entire recording was misclassified. It is that a brief temporal error can change the semantic description of an otherwise well-recognized procedure.

The plotted rows use common centers, so narrow gaps reflect the sampling and validity protocol. The adaptive algorithm uses the denser stored stream and therefore cannot be reconstructed merely by measuring the widths of these sparse colored blocks. The segment audit in Figure 21 provides the corresponding dense-stream evidence. This distinction prevents a visually convenient timeline from silently replacing the actual decoder input.

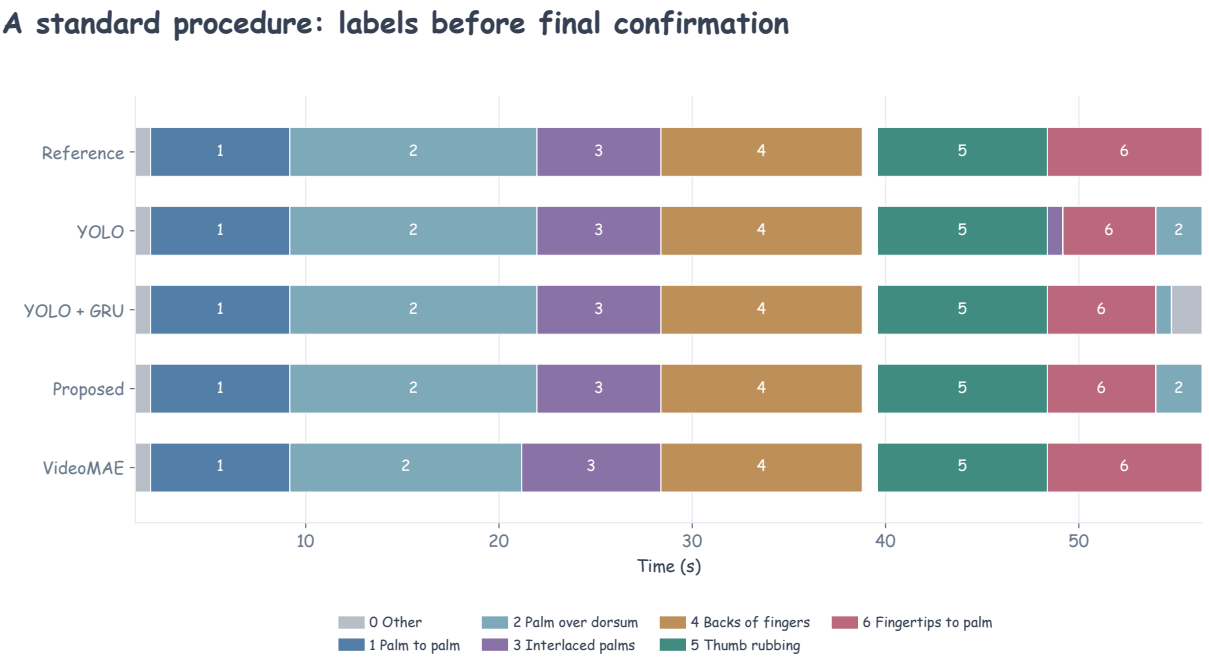


Figure 18: Common-center timelines for supplemental correct recording 04. The proposed row shows class output before final adaptive confirmation, making the extra terminal candidate visible.

10.2 A reversed and reordered procedure

The nonstandard recording identified by 645321 tests whether the model can preserve a deliberately incorrect order. Figure 19 shows that the observed sequence is not coerced into the standard pattern. All four retained pipelines match this particular reference chain, although their fine-grained transition positions are not identical.

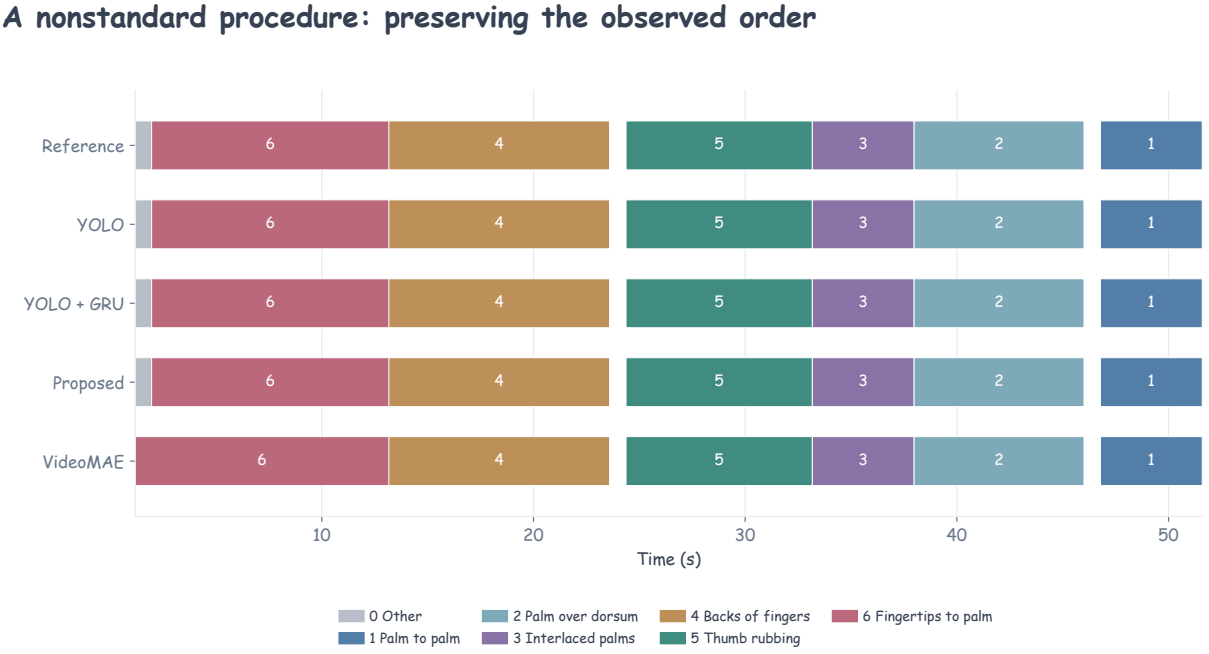


Figure 19: Common-center predictions for the nonstandard 645321 recording. The observed order is retained; it is not converted to the standard six-step sequence.

This successful case should not be used to imply that every model handles every reordered recording equally well. The all-video matrix shows failures on other nonstandard sequences. Its role is more specific: it demonstrates the type of evidence needed to distinguish faithful recognition from a hard-coded normal-order decoder. The assessment layer subsequently compares the recognized chain with 123456 and can correctly identify the procedure as out of order.

The same logic applies to omissions. A chain such as 12356 should remain missing Step 4 if that is what the recording shows. Adding the absent action to satisfy a procedural template would be an error. This is why the report includes nonstandard and incomplete reference chains in its primary sequence analysis rather than evaluating only correctly performed demonstrations.

10.3 Original Video 8 and the Step 4/5 boundary

Original Video 8 is particularly useful because it combines a reviewed label correction with an adaptive-chain correction. Its final reference is 124536. Figure 20 focuses on the region around the Step 4/5 transition using dense predictions. The annotation places Step 5 at 29.0 seconds after an uncertain interval from 28.6 seconds. The upper panel shows class labels, and the lower panel shows the stored winning-label confidence for the three YOLO-family pipelines.

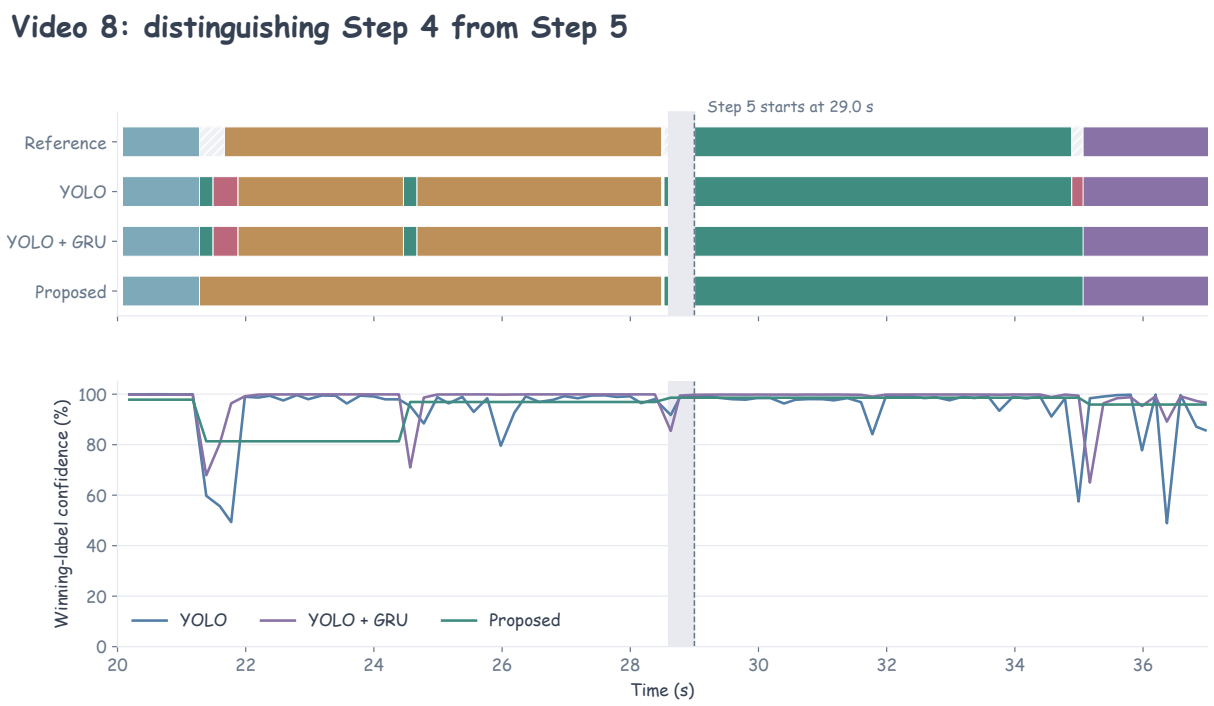


Figure 20: Dense class and winning-confidence traces around the reviewed Step 4/5 transition in original Video 8. The vertical marker indicates the annotated Step 5 start at 29.0 seconds.

The confidence lines should not be read as fixed-class probabilities. When a model changes its winning class, the plotted number refers to the newly winning class. A high trace therefore does not prove that the model is confident in Step 4 throughout the interval, nor does a dip identify the boundary-head output. The class timeline and reference annotation are needed to interpret the confidence trace correctly.

The corrected interval resolves a source of disagreement in earlier interpretations, but it does not explain every later sequence error. The proposed raw chain 1245346 includes an additional Step 4 after Step 3. That later candidate is addressed by adaptive confirmation, yielding 124536.

The report separates annotation correction from algorithmic correction because they operate on different sides of the evaluation: one changes the reference record, while the other changes the predicted chain.

10.4 Inspecting actual confirmation decisions

Figure 21 plots the segment-level evidence for original Video 8 and supplemental correct recording 04. The horizontal axis is the robust log-duration score, while the vertical axis is segment confidence minus the recording's median action confidence. A point in the lower-left rejection region is both unusually short and below typical confidence. The red crosses identify the segments left unconfirmed.

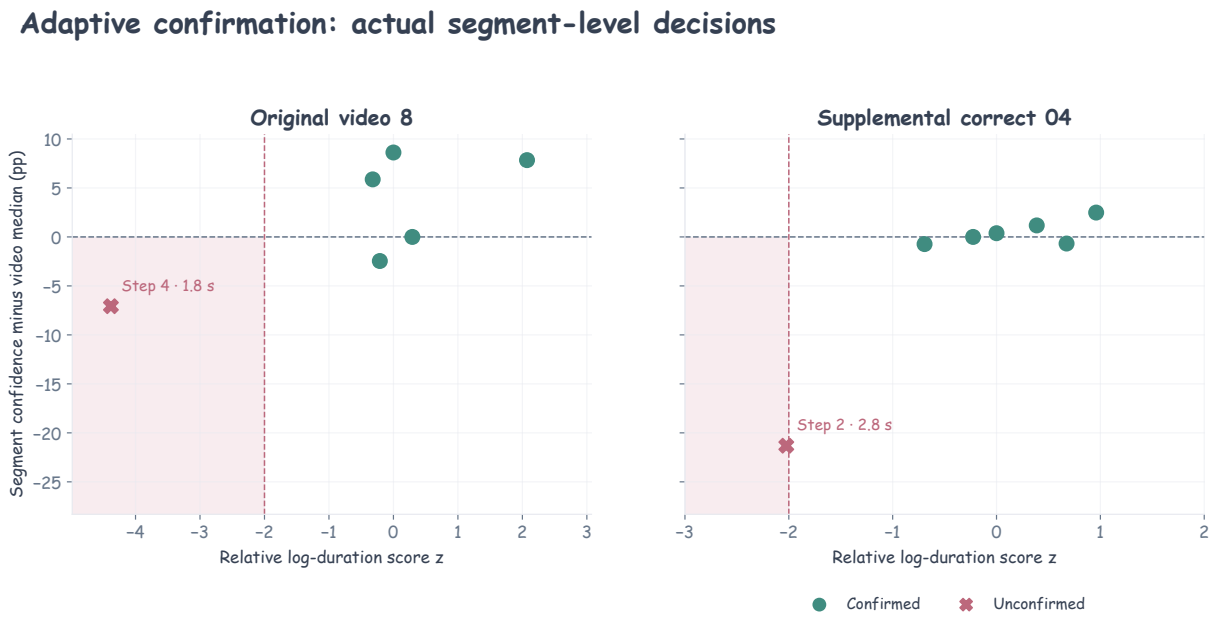


Figure 21: Actual dense-stream segment decisions for the two corrected chains. Red crosses satisfy both the relative-duration and confidence conditions; other candidates remain confirmed.

For original Video 8, the additional Step 4 lasts 1.8 seconds in the sampled decoder representation. Its duration score is approximately -4.38 and its available mean class confidence is about 82.92%. The latter is below the recording's median action confidence, so both conditions hold. The relevant fact is not that 82.92% is universally low; it is weaker than the typical action segment in that recording and accompanies a marked duration outlier.

For supplemental correct recording 04, the extra Step 2 lasts 2.8 seconds and also lies in the joint rejection region. Its removal explains the change from 1234562 to 123456. A fixed three-second rule would reject both examples as well, but that observation would not distinguish the new algorithm. The adaptive rule's defining behavior is that duration is assessed relative to the recording and combined with confidence, allowing other short but well-supported segments to remain eligible.

These case studies establish what the retained pipeline did on two actual recordings. They do not establish a universal minimum action duration, nor do they prove that all genuine brief actions will survive. The complete segment records, including confirmed and unconfirmed intervals, are retained so that this trade-off can be evaluated on future examples rather than concealed behind the final chain alone.

11 Camera Application and Deployment

11.1 One selected model per session

The camera application turns the recognition pipeline into a complete session workflow. Before acquisition, the user chooses a camera and one of the available models. The selected model alone processes that session. This avoids presenting four competing results as though they were one live assessment and makes the computational workload easier to interpret. Model selection is locked once the session begins so that one chain does not mix predictions from different methods.

Figure 22 shows the delivered interface before acquisition. Its layout separates model setup, camera preview, current action and duration assessment. The current-action area displays a provisional label and confidence. The sequence indicators summarize the action progression, while the duration panel shows recognized washing time against the configured threshold. The completion criteria are visible rather than being revealed only after the recording ends.

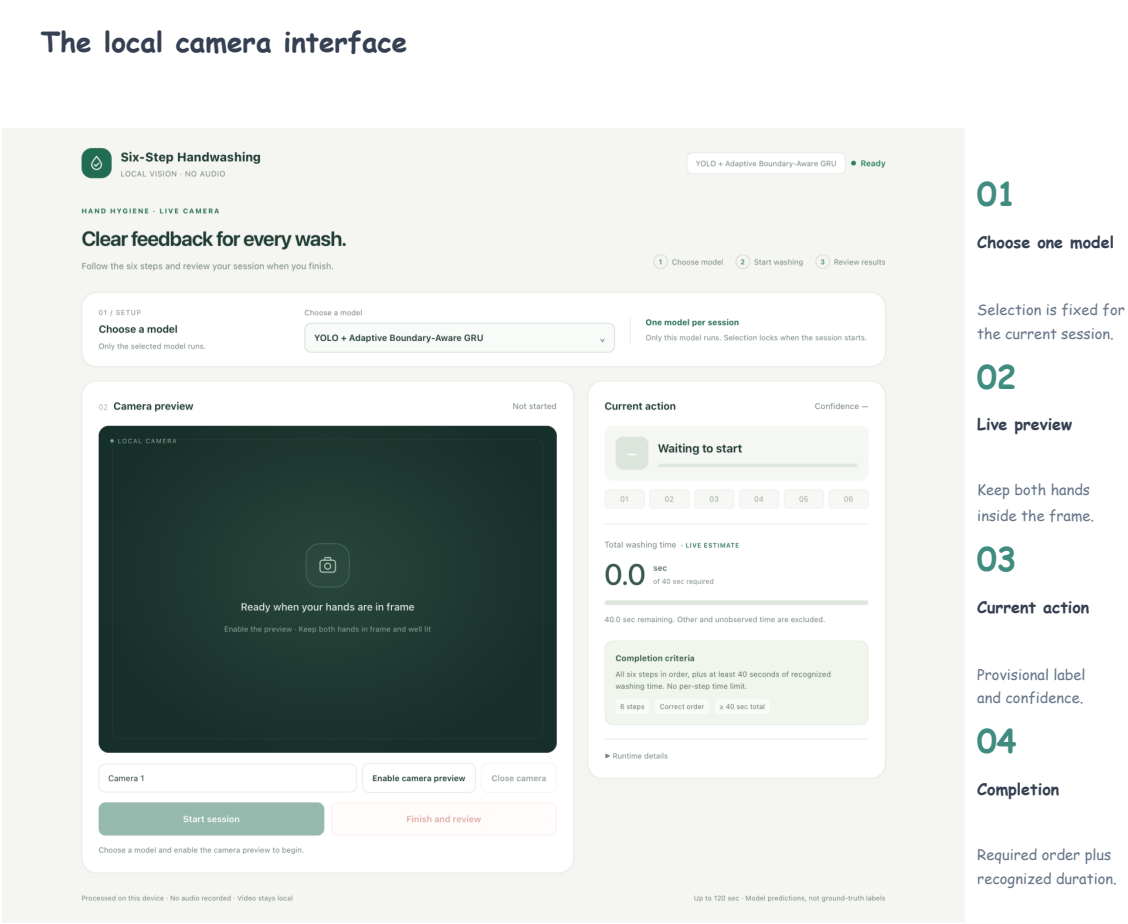


Figure 22: The local camera interface before a session starts. The controls support model selection, camera preview and session review; the displayed duration criterion is part of the application configuration.

The image is a real interface screenshot, not a simulated successful prediction. A blank preview and zero duration are appropriate before acquisition. The report does not replace them with invented scores to make the system appear active. The accompanying architecture and sequence diagrams explain what happens after the user starts a session.

Sessions are limited to 120 seconds. Audio is not acquired. Sampled images and session results remain local, supporting later inspection of the same recording when a prediction is questioned.

These properties define the implemented interaction; they are not a claim that every possible deployment environment has the same storage or security configuration.

11.2 Incremental inference and cached finalization

A central usability objective is to avoid requiring the user to wait for the entire recording to be processed again after washing. The application therefore performs incremental work during acquisition and caches its outputs. Figure 23 separates the browser, frame queue, selected model and result cache. Finishing the session stops the admission of new frames, drains already accepted work and then finalizes the report from the cached sequence.

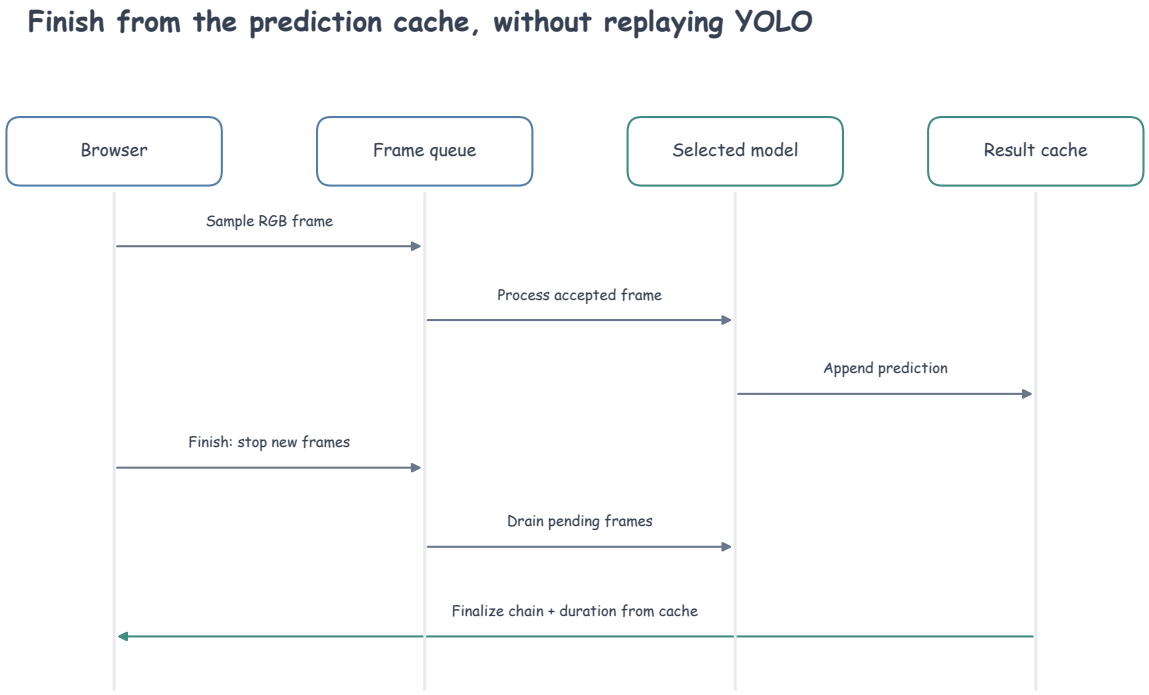


Figure 23: Session finalization from cached predictions. Finishing stops new frame admission, waits for accepted work and constructs the result without replaying the recording through YOLO.

Cached finalization does not imply that every final result appears with zero delay. If inference is slower than acquisition, accepted frames may still be pending. Their completion is necessary to avoid reporting a chain that omits the end of the procedure. The engineering objective is to eliminate unnecessary repeated work and keep the outstanding queue bounded; the final response time still depends on the actual workload and device.

The live and finalized outputs also have different evidential status. A current label can be produced from the available frame and recurrent state, while confirmation statistics evolve as more action segments appear. Earlier decisions can change when the full sequence becomes available. The interface therefore describes live feedback as provisional. This is consistent with a unidirectional recurrent backbone combined with recording-level final confirmation.

A session cache should retain timestamps and model identity with the predictions. Otherwise, a visually plausible final chain could be associated with an old recording or the wrong selected model. Clearing the displayed result at the start of a new session and maintaining a distinct session

lifecycle are part of making the output interpretable. An apparently stale interface is a systems problem even when the model weights are correct.

11.3 Recognized washing duration and completion

Duration assessment operates on disjoint timestamped intervals. Let $[s_i, e_i)$ be an observed interval and y_i its recognized class. The amount of recognized washing is

$$T_{\text{wash}} = \sum_i (e_i - s_i)\, \mathbf{1}[1 \le y_i \le 6] \tag{23}$$

Other intervals do not contribute. Time without an available prediction is tracked as unknown rather than silently counted as washing. The assessment uses observed intervals clipped to the recording extent, which prevents elapsed wall-clock time from being mistaken for recognized action time.

The application's completion rule is

$$\text{Complete} = \mathbf{1}[\mathcal{C} = (1, 2, 3, 4, 5, 6)] \wedge \mathbf{1}[T_{\text{wash}} \ge 40 \text{ s}] \tag{24}$$

There is no required minimum duration for each individual step. The total-time requirement and the chain requirement are evaluated separately. A short interval may contribute to washing time while remaining unconfirmed as a distinct step, because recognizing washing activity and identifying a separate action instance are different decisions.

Unknown time also affects the wording of the result. If recognized washing is already sufficient, the duration criterion is met. If recognized washing plus all unknown time would still be below the threshold, the criterion is not met. If the threshold could be reached only by assuming that unknown time was washing, the evidence is insufficient to confirm duration. The application can then request review instead of pretending that missing predictions are observed background or observed washing.

The output should explain missing steps, incorrect order and insufficient time in understandable terms. It should not label an unannotated live recording with an accuracy percentage. Confidence is a probability associated with a prediction; accuracy requires comparison with a reference annotation. This distinction remains important when the displayed confidence is high and a user notices an incorrect gesture.

11.4 Local model cost and temporal waiting

Figure 24 compares the retained architecture sizes and historical local forward-pass measurements. The proposed YOLO/boundary architecture contains 10,642,639 parameters, while VideoMAE contains 86,232,583. The ratio is approximately 8.10. This is a direct model-size distinction, but parameter count alone does not measure battery use, throughput or all memory consumed by the application.

The historical measurements use MPS, batch size one and three synthetic repetitions. Their medians are 148.63 milliseconds for the proposed frame/update unit and 317.58 milliseconds for a VideoMAE window. The VideoMAE timing uses a randomly initialized model with the retained architecture configuration. These numbers describe short forward-pass measurements with different input units. They do not establish a sustained end-to-end speed ratio, and cumulative process memory is not reported as though it were isolated per-model memory.

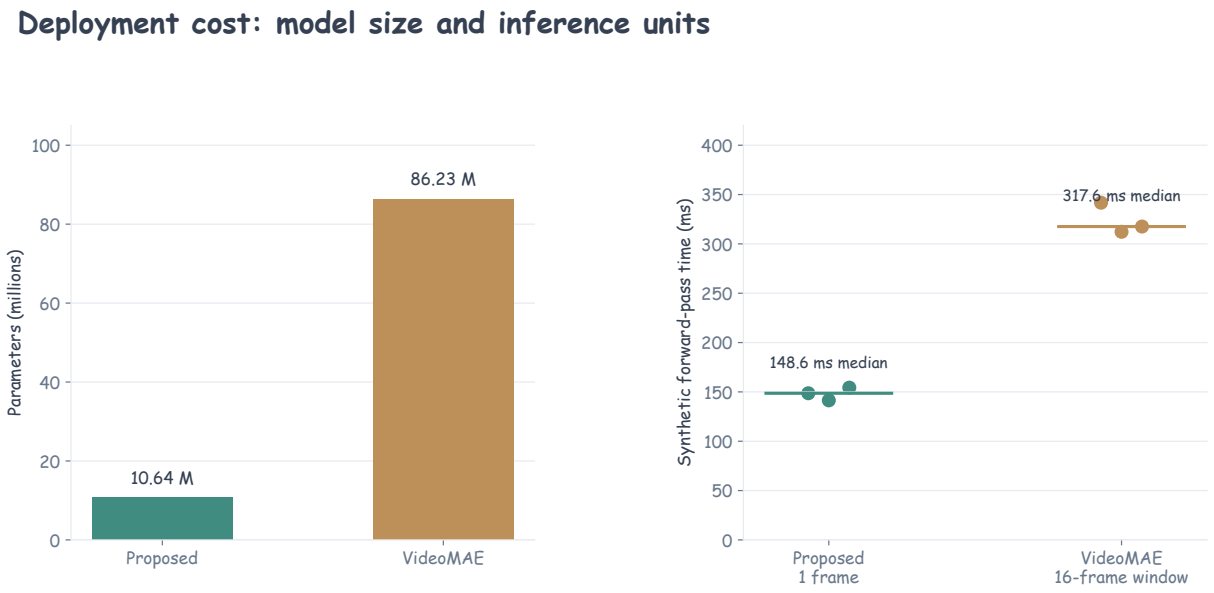


Figure 24: Parameter counts and three historical synthetic MPS timing samples. The proposed input unit is one frame with a temporal update; the VideoMAE unit is a 16-frame window.

VideoMAE can produce streaming predictions, but window collection and target timing introduce additional latency. The first 16-sample window spans approximately three seconds at 5 fps, and the centered prediction uses about 1.4 seconds of future context before computation. A recurrent image-feature path can produce an initial class update before such a window has been collected. The full proposed chain still benefits from later segment evidence, so immediate class feedback and final procedural assessment must remain distinct claims.

These observations support retaining the proposed method as the camera-oriented choice and VideoMAE as the strongest internal accuracy reference. They do not justify saying that VideoMAE is impossible to deploy locally. It can run on the local computer; its larger model and delayed window contract create different practical trade-offs. A deployment recommendation is more credible when it states those trade-offs precisely rather than dismissing the stronger recognizer.

11.5 Operational interpretation

Camera placement remains part of the input conditions. Both hands should be visible, adequately illuminated and large enough in the image to resolve the contact pattern. An unusual viewpoint can make a familiar gesture resemble another class, potentially with high confidence. That behavior does not automatically establish a mapping bug, but a reported error should still be checked against the saved class contract and the corresponding frame before attributing it to viewpoint alone.

The delivered runtime uses MPS when available and CPU fallback otherwise. A successful installation therefore requires compatible runtime libraries and the expected model assets, not just the front-end HTML page. The dashboard is the visible entry point to a local inference service. The package's quick-start and model manifests connect that interface to the backend and weights so that the technical result can be reproduced as a working system.

12 Discussion, Limitations and Next Evaluation

12.1 What the boundary-aware pipeline contributes

The strongest observed improvement within the YOLO family is the quality of the temporal output. The proposed method reduces extra output boundaries relative to ordinary GRU and produces ten matching final chains. Its class accuracy improves modestly, while its macro F1 remains slightly below standalone YOLO. This pattern supports a specific conclusion: the delivered boundary-aware and confirmation pipeline is useful for reconstructing procedures in the reviewed scene, but it is not a universal improvement in every aspect of recognition.

The term boundary-aware should therefore be connected to observable behavior rather than treated as an automatic quality guarantee. The learned branch, segment decoder and adaptive confirmation all contribute to the configured system. The available evidence compares that system with its alternatives. It does not isolate the causal contribution of each component. In particular, the proposed final chain uses denser predictions than the common-center raw chains, so the final-chain improvement cannot be assigned entirely to the duration rule.

The confusion matrices also show where temporal consistency can come at a cost. Background evidence can be absorbed into a dominant action segment, and a persistent but incorrect label can survive local smoothing. These weaknesses are relevant to both action time and chain interpretation. A future refinement should preserve the good temporal structure while improving the treatment of Other, rather than focusing only on making the already-correct action chain look cleaner.

12.2 Choosing a primary model without changing the evidence

Different operational objectives can select different models. If the objective is the highest retained classification and boundary score, VideoMAE is the strongest choice. If the objective includes a compact local model, incremental frame feedback and a complete action-chain result, the proposed pipeline has a defensible role. It reaches the same ten-chain total with substantially fewer parameters and a recurrent input contract compatible with the camera workflow.

The recognition composite does not settle this decision because it omits deployment cost. Adding arbitrary efficiency weights solely to reverse a ranking would obscure the evaluation. A more transparent approach is to report the measured recognition results, the parameter count, the temporal waiting requirements and the available timing evidence separately. The selected model can then be justified against the intended application instead of being declared the winner of an unspecified universal score.

The simpler baselines remain valuable. Standalone YOLO demonstrates that the target appearances are largely recognizable and provides a useful check on class mapping and background behavior. Ordinary GRU demonstrates the benefit and limits of recurrent context. Neither is discarded as intrinsically unsuitable; their lower exact-chain results under the retained configuration explain why they are not the primary procedural-feedback method. They remain selectable and inspectable alternatives in the application.

12.3 Limits of the retained validation

The corpus contains substantial frame-level annotation coverage, including 6,987 sampled records, but those records originate from complete videos in a particular recorded scene. Their temporal correlation limits the interpretation of frame counts as independent evidence. The ten evaluation videos also have a development history: they were repeatedly inspected, and some labels were corrected after predictions had been viewed. The report therefore characterizes retrospective internal performance rather than unseen-person or unseen-viewpoint generalization.

There are additional protocol differences across models. VideoMAE uses generic pretraining and lower-resolution video windows. The recurrent heads are finally fit on all 22 development clips, while VideoMAE uses the retained 14/8 training-validation procedure. Their final chain postprocessing is not shared. These differences do not make the practical comparison meaningless, but they constrain architecture-only claims. A controlled study would need to hold the relevant factors fixed and define its target question before model selection.

The boundary metric is also limited by sampling and annotation precision. It measures output label changes against intervals with tolerance, not exact physical contact changes. The matched timing error excludes misses and extra events from its denominator. Reporting only that error could favor a system with undesirable segmentation behavior. Keeping the event counts, F1 and timing together is necessary for an honest account of temporal quality.

12.4 A focused independent evaluation plan

The next useful evaluation would begin by freezing the complete pipeline: class mapping, preprocessing, weights, normalization, decoder settings, confirmation settings and completion rule. New videos should then be collected and annotated without using their model predictions to tune those choices. Standard procedures, omissions, repetitions, reordered actions, background periods and genuinely short actions should all be represented. Recording conditions should deliberately cover the viewpoints and lighting expected in actual use.

The unit of sequence evaluation should remain the complete recording. Frame-level metrics can describe recognition, but video-level chain matching reveals whether the system reconstructs a whole procedure. Per-class support and confusion should be retained, especially for Other and visually similar gestures. Boundary counts and interval definitions should be documented with the same care as in the current comparison. If confidence calibration is studied, complete probability vectors should be saved rather than only winner confidence.

A separate runtime evaluation should measure a sustained camera session on the intended device. Useful observations include admitted sampling rate, processed-frame rate, queue delay, first-label delay and time from finishing to the final result. The processing unit must be explicit when comparing a recurrent frame update with a video window. Model size, isolated memory, power and throughput answer different questions and should not be inferred from a single short timing measurement.

This plan extends the evidence without requiring the report to promise new performance. The current deliverable is a technically complete local prototype with inspectable internal results. Independent recordings and matched deployment measurements would determine how broadly its present advantages persist.

12.5 Reproducibility as part of the contribution

The release includes model contracts, annotation records, split manifests, saved predictions, metric code and checksums. The accompanying figure collection adds aligned prediction tables, confusion matrices and segment-decision records. These assets allow a reviewer to move from a reported percentage to the observations and definitions that produced it. They also make it possible to correct a reference annotation transparently without silently retaining obsolete scores elsewhere in the report.

The report, application and presentation serve different purposes. The report explains assumptions and evidence; the application demonstrates the interaction; the presentation communicates the main technical decisions within a limited time. A polished interface does not substitute for evaluation, and an extensive report does not substitute for a functioning session workflow. The delivered project is strongest when these artifacts agree about model identity, timing rules and the meaning of a successful result.

13 Conclusions

The project delivers a camera-based handwashing assessment pipeline that connects image recognition to temporal segmentation, action-chain reconstruction and recognized-duration assessment. The initial PSKUS study supports the use of temporal context and transition-aware reasoning. The target-scene corpus then supplies 6,987 sampled annotation records covering standard procedures, nonstandard orders, omissions and isolated gestures.

The proposed YOLO + Adaptive Boundary-Aware GRU combines a frozen visual representation, recurrent context, learned boundary evidence and relative duration-confidence confirmation. Its confirmation rule replaces a fixed three-second action admission threshold while preserving the order actually observed. On the retained ten-video comparison, it achieves 96.52% accuracy, 93.71% macro F1 and 94.40% output-boundary F1, with all ten final chains matching their references. The case studies show how two brief extra candidates are left unconfirmed and why that changes the complete chain without implying perfect frame labels.

VideoMAE remains the strongest recognition reference, achieving 99.53% accuracy and the same ten-chain result. The proposed method is selected for the local camera workflow because it combines that final-chain outcome with approximately 10.64 million parameters and an incremental recurrent path. This is a deployment-oriented choice supported by the retained evidence, not a claim that the model exceeds VideoMAE on every metric.

The application runs one chosen model per session, presents provisional feedback and finalizes from cached predictions after accepted frames complete. Its completion rule requires the standard six-step order and at least 40 seconds of recognized washing, with Other and unknown time treated separately. The delivered data, weights, configurations and prediction records make the result inspectable and reproducible.

The findings establish retrospective internal performance in the reviewed scene. The next evidential step is a frozen-pipeline evaluation on independently collected recordings and sustained runtime measurements on the target device. Within its stated scope, the project provides a complete technical path from annotated video to practical local procedural feedback.

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A Complete Predicted Action Chains

The following tables retain all forty final chains from the four-model comparison. Standard and nonstandard recordings are separated to make the intended reference behavior clear. Digits denote Steps 1–6; Other is removed before chain comparison. A repeated digit separated by a different action represents a repeated predicted step, not a typographic error.

The proposed entries are the finalized dense-stream chains after adaptive confirmation. The other entries are their archived raw common-center chains. These tables preserve the same protocol as the main result table. They should not be interpreted as a comparison in which identical postprocessing has been applied to every model.

Table 7: Final predicted chains for standard procedures

Video	Model	Reference	Predicted final chain	Match
Original 4	YOLO	1-2-3-4-5-6	1-2-6-1-3-1-3-1-3-4-5-6-4-6-4	No
Original 4	YOLO + GRU	1-2-3-4-5-6	1-2-6-3-4-5-6	No
Original 4	Proposed	1-2-3-4-5-6	1-2-3-4-5-6	Yes
Original 4	VideoMAE	1-2-3-4-5-6	1-2-3-4-5-6	Yes
Correct 01	YOLO	1-2-3-4-5-6	1-2-3-4-5-6-5-6	No
Correct 01	YOLO + GRU	1-2-3-4-5-6	1-2-3-4-5-6	Yes
Correct 01	Proposed	1-2-3-4-5-6	1-2-3-4-5-6	Yes
Correct 01	VideoMAE	1-2-3-4-5-6	1-2-3-4-5-6	Yes
Correct 02	YOLO	1-2-3-4-5-6	1-2-3-4-5-4-6	No
Correct 02	YOLO + GRU	1-2-3-4-5-6	1-2-3-4-5-6	Yes
Correct 02	Proposed	1-2-3-4-5-6	1-2-3-4-5-6	Yes
Correct 02	VideoMAE	1-2-3-4-5-6	1-2-3-4-5-6	Yes
Correct 03	YOLO	1-2-3-4-5-6	1-3-2-3-4-5-6-3-6	No
Correct 03	YOLO + GRU	1-2-3-4-5-6	1-3-2-3-4-5-6	No
Correct 03	Proposed	1-2-3-4-5-6	1-2-3-4-5-6	Yes
Correct 03	VideoMAE	1-2-3-4-5-6	1-2-3-4-5-6	Yes
Correct 04	YOLO	1-2-3-4-5-6	1-2-3-4-5-3-6-2	No
Correct 04	YOLO + GRU	1-2-3-4-5-6	1-2-3-4-5-6-2	No
Correct 04	Proposed	1-2-3-4-5-6	1-2-3-4-5-6	Yes
Correct 04	VideoMAE	1-2-3-4-5-6	1-2-3-4-5-6	Yes

For nonstandard recordings, a prediction matches when it reproduces the nonstandard reference faithfully. Such a match is compatible with an incomplete procedural verdict. Original Video 8 uses the corrected reference 124536, including Step 5 starting at 29.0 seconds in its frame-level annotation.

Table 8: Final predicted chains for nonstandard procedures

Video	Model	Reference	Predicted final chain	Match
Original 8	YOLO	1-2-4-5-3-6	1-3-2-4-5-3-4-6-4-6	No
Original 8	YOLO + GRU	1-2-4-5-3-6	1-3-2-4-5-3-4-6-4-6	No
Original 8	Proposed	1-2-4-5-3-6	1-2-4-5-3-6	Yes
Original 8	VideoMAE	1-2-4-5-3-6	1-2-4-5-3-6	Yes
Nonstandard 12356	YOLO	1-2-3-5-6	1-2-3-5-6-3-6	No
Nonstandard 12356	YOLO + GRU	1-2-3-5-6	1-2-3-5-6	Yes
Nonstandard 12356	Proposed	1-2-3-5-6	1-2-3-5-6	Yes
Nonstandard 12356	VideoMAE	1-2-3-5-6	1-2-3-5-6	Yes
Nonstandard 12564	YOLO	1-2-5-6-4	1-2-5-6-4	Yes
Nonstandard 12564	YOLO + GRU	1-2-5-6-4	1-2-5-6-4	Yes
Nonstandard 12564	Proposed	1-2-5-6-4	1-2-5-6-4	Yes
Nonstandard 12564	VideoMAE	1-2-5-6-4	1-2-5-6-4	Yes
Nonstandard 32654	YOLO	3-2-6-5-4	3-1-3-1-3-2-6-5-4	No
Nonstandard 32654	YOLO + GRU	3-2-6-5-4	3-2-6-5-4	Yes
Nonstandard 32654	Proposed	3-2-6-5-4	3-2-6-5-4	Yes
Nonstandard 32654	VideoMAE	3-2-6-5-4	3-2-6-5-4	Yes
Nonstandard 645321	YOLO	6-4-5-3-2-1	6-4-5-3-2-1	Yes
Nonstandard 645321	YOLO + GRU	6-4-5-3-2-1	6-4-5-3-2-1	Yes
Nonstandard 645321	Proposed	6-4-5-3-2-1	6-4-5-3-2-1	Yes
Nonstandard 645321	VideoMAE	6-4-5-3-2-1	6-4-5-3-2-1	Yes

B Data Contracts and Reproducible Evaluation

B.1 Frame and prediction records

A frame-level annotation record associates a video identity, sampled image, timestamp, class target and validity flag. Some historical folders use different column names for these concepts. The analysis resolves those schema variants explicitly rather than relying on their column order. A blank or invalid target does not become a new class, and an uncertain transition is not silently converted to Other.

Dense prediction exports include video identity, sample index, actual timestamp, ground truth, evaluation validity, predicted label and winning-label confidence for each YOLO-family method. VideoMAE exports its window-center time, reference label, predicted class and confidence. The aligned evaluation joins these records by video and exact timestamp. A record is accepted only when its reference label and validity are consistent with the common-center protocol.

The final result file stores aggregate metrics and per-video chains. Figure generation recomputes confusion matrices and class metrics from the saved predictions and checks their agreement with the archived aggregates. The segment audit separately uses the dense proposed predictions. Keeping those two input paths distinct prevents an accidental change in sampling density from being hidden inside a visualization.

B.2 An auditable analysis sequence

A reproducible review can proceed in a fixed order. First, identify the final dataset and model manifests. Second, verify the relevant source hashes and load the corrected annotation records. Third, build the common-center set using exact times and valid references. Fourth, calculate class confusion and output-boundary matches. Fifth, construct raw chains on the common centers and obtain the proposed final chain from its dense decoder. Finally, compare each chain with its complete reference and retain the intermediate outputs.

The intermediate records matter when a result changes. A class-mapping error, annotation correction, timestamp mismatch or decoder change can each affect a score differently. Saving only the final percentage makes these causes difficult to distinguish. Saving the aligned rows, boundary events and segment decisions allows a reviewer to identify which stage changed and whether the new result belongs to the same protocol.

B.3 Figure provenance

The report's measured figures come from the retained annotations, prediction tables, result JSON and local architecture timing record. The historical PSKUS figure comes from the earlier project comparison. The six-action panel and system interface use supplied images. Architecture, context, boundary-matching and session-sequence diagrams illustrate the implemented method or protocol; their example positions and output chains are not additional observations.

The companion atlas contains SVG, PDF and PNG versions of the figures, CSV tables and LaTeX table fragments. PDF figures preserve vector text and geometry for report composition. Photographic content remains raster even when it is embedded in a vector file. Source snapshots and their hashes allow the plotted evidence to be traced back to the retained records [8].

C Deployed Configuration and Session Contract

Table 9: Retained settings of the deployed temporal pipeline

Component	Setting	Purpose
Input	1088 × 1088; approximately 5 fps	Match the archived visual extractor
Visual feature	1280 dimensions; frozen YOLO	Shared learned image representation
Temporal encoder	Projection 128; unidirectional GRU 128	Track recurrent temporal context
Heads	7-class output; boundary MLP 256 → 64 → 1	Recognize actions and candidate changes
Training loss	CE + 0.35 BCE + 0.5 interval loss	Archived final boundary-head objective
Boundary decoder	Threshold 0.3; peak gap 4 samples	Create candidate segments
Label smoothing	1 second, odd centered window	Suppress local class flicker
Adaptive confirmation	Log-duration z < -2 AND below-median confidence	Leave atypical weak segments unconfirmed
Comparison minimum	At least 4 non-background segments	Avoid unstable duration comparisons
Completion rule	Standard six-step chain AND ≥ 40 s recognized washing	Exclude Other and unobserved time

Table 9 lists the principal settings of the retained temporal pipeline. The minimum peak gap is measured in samples, while the one-second smoothing window is converted to an odd number of samples at the specified nominal rate. The duration-outlier cutoff is dimensionless. These quantities should not be described as if they were all fixed action-duration requirements.

Model assets bind weights to preprocessing, class contracts and normalization. The boundary model also requires its decoder settings, while VideoMAE requires its image-processor configuration. Changing these contracts changes the executed system.

C.1 Session and result contract

A session proceeds through setup, preview, acquisition, finishing and cached finalization. New sessions reset the model state and displayed evidence. Finishing stops new frames but waits for accepted work. Model identity and timestamps remain attached to the current session so that cached output cannot be mistaken for a new recording.

The final chain contains confirmed observed steps. Recognized washing time includes intervals assigned to the six actions; missing prediction time remains unknown. Completion requires the standard order and sufficient recognized washing time. A result describes visible procedure and does not certify cleanliness or live accuracy.

C.2 Technical delivery

The full release contains the application, weights, data, annotations, predictions and provenance. Source and documentation are hosted in the repository; larger assets accompany the complete delivery. This Overleaf project contains the report sources and vector figures and compiles independently of the model runtime.